Cheatsheet · 17 chapters on one page

Computer Vision cheatsheet

Every formula, key idea, table, memory aid and common mistake from the chapters, gathered in one place for quick revision. Nothing here is new. Each card links back to the lesson that teaches it, so open the lesson when a line does not make sense yet.

85 key ideas | 140 formulas | 86 tables | 50 tips and traps

STEP 1 Seeing like a machine

1.1 What computer vision is and why it is hard

The definition, the recognition tasks, why vision is hard, robustness and generalisation, the cost of images and video, and where computer vision sits in AI.

KEY IDEAS

- **The definition** Computer vision computes properties of the physical world from images and video. Its three jobs are measurement, perception and search.
- **The task family** Classification, detection (what and where), categorisation, instance recognition, scene, attribute, activity and relationship recognition, and segmentation.
- **Why it is hard** Illumination, viewpoint, scale, occlusion, clutter, intra class variation, inter class similarity, field of view, weather and messy environments.
- **Robustness and generalisation** Robustness handles expected variation. Generalisation handles new or shifted data. Imbalance hides poor performance on rare cases.
- **Size and cost** N = H × W × C, bytes = N × b ÷ 8, levels = 2 to the power b, raw video rate = frame bytes × fps. Systems trade accuracy against time, memory and energy.
- **Classical and learned features** Classical features are designed by hand and are cheap and interpretable. Deep networks learn features in a hierarchy from edges to objects.

FORMULAS

Normalised cross correlation (NCC)

$$\mathrm{NCC} = \frac{\Sigma\,(T-\bar{T})(I-\bar{I})}{\sqrt{(\,\Sigma\,(T-\bar{T})^2 \cdot \Sigma\,(I-\bar{I})^2\,)}}$$

T is the template, I is the image patch, and a bar means the average. The score runs from −1 to 1. If $I = a \cdot T + b$ with $a > 0$, the score is exactly 1, so brightness and contrast changes do not matter.

Lesson: Why vision is hard

Numbers in an image Exam favourite

$$N = H \times W \times C$$

H rows, W columns, C channels (1 for grey, 3 for colour).

Lesson: Images, video and their cost

Bytes in an image

$$\text{bytes} = \frac{H \times W \times C \times b}{8}$$

b is the bits per number. With 8 bits each number is one byte.

Lesson: Images, video and their cost

Levels per number

$$\text{levels} = 2^b$$

8 bits give 256 levels, from 0 to 255.

Lesson: Images, video and their cost

Raw video data rate

Exam favourite

$$\text{rate} = H \times W \times C \times \frac{b}{8} \times \text{fps}$$

Bytes per second before compression. fps is frames per second.

Lesson: Images, video and their cost

TABLES

Table: job · the question it answers · everyday example

Job	The question it answers	Everyday example
Measurement	How big, how far, what shape, how fast?	A phone measuring a room, a car estimating the distance to the car in front
Perception and interpretation	What is in the scene and what is happening?	Face unlock, a doctor's tool marking a tumour, a robot finding a cup
Search, organisation and interaction	Which pictures match, and what do they mean?	Searching your gallery for "beach", reverse image search, answering a question about a photo

Lesson: What computer vision is

Table: task · question · output

Task	Question	Output
Image classification	What is this picture mainly of?	One label, for example "dog"
Object detection	What objects are present, and where?	Boxes with a label and a confidence each
Categorisation	Which general class is this object?	Person, vehicle, chair
Instance recognition	Which exact object is this?	"This is the Eiffel Tower", "this is your car"
Scene recognition	What kind of place is this?	Kitchen, highway, market, forest
Attribute recognition	What is it like?	Red, metal, wet, broken, open
Activity recognition	What is happening over time?	Walking, falling, crossing the road
Relationship recognition	How do the objects relate?	"A person is holding a bag", "a car is approaching a cyclist"
Segmentation	Which pixels belong to which thing?	A label for every pixel

Lesson: What recognition involves

Table: challenge · what happens · example

Challenge	What happens	Example
Illumination	The same object gives different numbers under different light	Night, glare, shadows, a face lit from one side
Viewpoint and pose	The shape on the image changes with the camera angle	A car from the front and from the side
Scale	Objects far away are only a few pixels	A pedestrian 100 metres away
Occlusion	Part of the object is hidden	A child standing behind an adult
Clutter	A busy background hides boundaries	A crowded market, a supermarket shelf
Intra class variation	Members of one class look different	Office chair, stool, wheelchair
Inter class similarity	Different classes look alike	Wolf and husky, bus and truck
Limited field of view	Important things are outside the frame	A car coming from behind the camera
Weather	Contrast and visibility change	Rain, fog, dust, snow
Irregular environments	The world is messier than the dataset	Potholes, missing lane markings

Lesson: Why vision is hard

Table: cost · question to ask · why it matters

Cost	Question to ask	Why it matters
Time	Can each frame be processed before the next one arrives?	A car at 72 km/h moves 2 metres in 100 ms
Memory	Can the frames and the model fit on the device?	Phones, drones and cameras have little memory
Energy	How much battery or power does it use?	Edge devices run on small batteries

Lesson: Images, video and their cost

Table: era · how features were made · example

Era	How features were made	Example
Classical vision	People design the features by hand, then a simple classifier decides	SIFT or HOG features with a support vector machine
Deep learning	The network learns the features from labelled data	CNNs such as AlexNet and ResNet
Foundation models	Very large models learn from huge, loosely labelled data, often with text	CLIP, SAM, vision transformers

Lesson: Where computer vision sits in AI

TIPS AND TRAPS

REMEMBER — A picture is only a measurement of the world. The same object can produce very different numbers, and different objects can produce similar numbers. Every hard part of vision comes from this.

Lesson: What computer vision is

IN THE EXAM — Examiners like the pair detection versus recognition. Detection finds and localises objects. Recognition decides what they are and what they mean. Categorisation gives a class, instance recognition names one particular object.

Lesson: What recognition involves

COMMON MISTAKE — A high score on the test set does not prove the model will work after deployment. If the deployment conditions are missing from the data, the test set cannot measure them.

Lesson: Why vision is hard

COMMON MISTAKE Do not say that classical features such as HOG are needed "because CNNs overfit". The notes correct this. Overfitting depends on data size, model size, regularisation and augmentation. Classical features are useful because they are efficient, simple and easy to interpret.

Lesson: Where computer vision sits in AI

COMMON MISTAKE OpenCV loads colour images in BGR order, not RGB. Most other libraries (Matplotlib, PIL, PyTorch models) expect RGB. Forgetting to convert is one of the most common bugs in vision code.

Lesson: Images in Python

1.2 Marr's theory, edges and hand made features

Marr's stages and the 2½-D sketch, what makes a good feature, image gradients and Sobel, SIFT keypoints and HOG descriptors.

KEY IDEAS

- **Marr's four stages** Input image, primal sketch (edges, blobs), 2½-D sketch (depth and orientation of visible surfaces), 3D model (full shape). 2½-D because it depends on the viewpoint.
- **Good features** Distinctive, stable (repeatable) and invariant. Corners beat edges, edges beat flat regions. Detection finds points, description turns them into vectors.
- **Gradients** Gx and Gy by differences or Sobel. Magnitude √(Gx² + Gy²), direction atan2(Gy, Gx), pointing across the edge from dark to bright.
- **SIFT** DoG scale space for scale, dominant orientation for rotation, 4 × 4 × 8 = 128 descriptor, ratio test d1 ÷ d2 < 0.8 for matching.
- **HOG** Cells of 8 × 8, 9 unsigned bins, votes weighted by magnitude, 2 × 2 block normalisation, 3,780 numbers for 64 × 128. Good for upright people, weak on pose changes and occlusion.
- **Why classical features matter** Efficiency, interpretability, domain knowledge and support for deep pipelines, not because CNNs overfit.

FORMULAS

Central difference

$$G_x \approx I(x+1, y) - I(x-1, y)$$

Often divided by 2. The kernel is [−1, 0, 1].

Lesson: Image gradients

Gradient magnitude Exam favourite

$$|\nabla I| = \sqrt{G_x^2 + G_y^2}$$

How strong the edge is.

Lesson: Image gradients

Gradient direction Exam favourite

$$\theta = \operatorname{atan2}(G_y, G_x)$$

Points across the edge, from dark to bright.

Lesson: Image gradients

Sobel kernels

$$K_x = \begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{pmatrix} \quad K_y = \begin{pmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{pmatrix}$$

Sobel is a central difference in one direction combined with a small smoothing [1, 2, 1] in the other direction, which makes it less sensitive to noise. Prewitt uses [1, 1, 1] instead.

Lesson: Image gradients

Gaussian blur

$$L(x, y, \sigma) = G(x, y, \sigma) * I(x, y)$$

G is a Gaussian of width σ and $*$ is convolution.

Lesson: SIFT: keypoints that survive zoom and rotation

Difference of Gaussians

$$\mathrm{DoG} = L(x, y, k\sigma) - L(x, y, \sigma)$$

Close to $(k-1)\,\sigma^2 \nabla^2 G * I$, the scale normalised Laplacian.

Lesson: SIFT: keypoints that survive zoom and rotation

Descriptor length

Exam favourite

$$4 \times 4 \text{ cells} \times 8 \text{ bins} = 128$$

Each cell is a small histogram of gradient directions.

Lesson: SIFT: keypoints that survive zoom and rotation

Lowe ratio test

$$\frac{d_1}{d_2} < 0.8$$

d_1 and d_2 are the distances to the nearest and second nearest descriptor. Accept the match only if the best is clearly better than the runner up.

Lesson: SIFT: keypoints that survive zoom and rotation

Bin width

$$\text{width} = \frac{180°}{\text{number of bins}}$$

9 bins give 20° per bin, with centres at 10°, 30°, 50° and so on.

Lesson: HOG: describing shape with edge directions

Vote split

$$\text{share to the upper bin} = \frac{\theta - c_{\text{lower}}}{\text{bin width}}$$

The rest goes to the lower bin. Both shares are multiplied by the magnitude.

Lesson: HOG: describing shape with edge directions

Descriptor length

Exam favourite

$$(n_x - 1)(n_y - 1) \times 4 \times B$$

For 64 × 128 with 8 × 8 cells: 7 × 15 = 105 blocks, each 4 × 9 = 36 numbers, so 3,780 in total.

Lesson: HOG: describing shape with edge directions

Block normalisation (L2)

$$v \leftarrow \frac{v}{\sqrt{\|v\|^2 + \varepsilon^2}}$$

ε is a small number that avoids dividing by zero.

Lesson: HOG: describing shape with edge directions

TABLES

Table: stage · what it contains · question it answers

Stage	What it contains	Question it answers
1. Input image	Pixel intensities, colour, local contrast	What signal did the camera capture?
2. Primal sketch	Edges, bars, blobs, ends, zero crossings, groups, boundaries	Where does structure begin and end?
3. 2½-D sketch	Depth and surface orientation of the visible surfaces, discontinuities	How are the visible surfaces arranged relative to me?
4. 3D model	Volumes and parts in an object centred frame	What is the object and what is its full shape?

Lesson: Marr's stages of vision

Table: property · meaning · bad example · good example

Property	Meaning	Bad example	Good example
Distinctive	Its neighbourhood looks unlike most other places	A patch of plain blue sky	The corner of a window
Stable (repeatable)	It is found again in a second photo of the same scene	A reflection that moves when you move	A painted letter on a sign
Invariant	Its description stays the same under scale, rotation and lighting changes	Raw pixel values	A normalised histogram of gradient directions

Lesson: What makes a good feature

Table: · sift · hog

	SIFT	HOG
Main use	Matching the same points across images	Describing the shape of a whole window for detection
Where it is computed	Around sparse keypoints, at their own scale	Densely, on a fixed grid of cells
Invariance	Scale and rotation by design, partly lighting	Lighting and small shifts, not rotation or pose
Length	128 per keypoint	3,780 for a 64 × 128 window

Lesson: HOG: describing shape with edge directions

Table: reason · what it means · example

Reason	What it means	Example
Efficiency	Little memory, little power, no training data needed	A doorbell camera, a microcontroller, a drone
Interpretability	You can say why the system fired	A HOG glyph shows the shoulder edges that made the detector fire
Domain knowledge	If you know the key pattern, you can measure it directly	Finding a logo, a QR code, the straight edges of a document
Working with deep models	They can prepare or support a learned model	Keypoints for camera tracking in AR, matching in 3D reconstruction

Lesson: Why classical features still matter

TIPS AND TRAPS

IN THE EXAM Depth is not volume. A depth map gives the distance to each visible surface. A 3D model describes the whole object, including the hidden back. The exam asks this as "why is it called 2½-D".

Lesson: Marr's stages of vision

IN THE EXAM Feature detection finds where the interesting points are. Feature description turns the neighbourhood of each point into a vector of numbers so points can be compared. SIFT does both. HOG is mainly a descriptor.

Lesson: What makes a good feature

COMMON MISTAKE With image coordinates the y axis points down. An angle of 90° therefore means "brightness increases downwards". Many textbooks draw y upwards, so check which convention a question uses.

Lesson: Image gradients

STEP 2 Images as numbers

2.1 Pixels, colour models and compression

Images as discrete signals, RGB, CMYK, HSL, YCbCr and BGR, lossy and lossless compression, JPEG and the DCT, and I, P and B video frames.

KEY IDEAS

- **Image as numbers** H × W × C values from 0 to 255. Check channel order, range, type, normalisation and layout before a model sees an image.
- **Colour models** RGB adds light, CMYK adds ink, HSL separates hue from lightness, YCbCr separates brightness from colour. OpenCV uses BGR. Grey = 0.299R + 0.587G + 0.114B.
- **Lossy and lossless** PNG keeps every pixel, JPEG throws away detail the eye misses. Use lossless for text, medical and legal images.

- **JPEG in five steps** YCbCr, chroma downsampling, 8 × 8 DCT, quantisation (the lossy step), zig zag plus Huffman coding. DC = 8 × mean.
- **I, P and B frames** I stands alone, P predicts from the past, B from past and future. B is smallest but adds delay, so live systems avoid it.
- **Compression and evidence** Edge processing and content aware compression save bandwidth, but heavy compression can erase important detail.

FORMULAS

Image as a function

$$I : \{0, \ldots, H-1\} \times \{0, \ldots, W-1\} \to \{0, \ldots, 255\}^C$$

Each position (row, column) maps to C numbers.

Lesson: An image is a discrete signal

Stored values

Exam favourite

$$H \times W \times C$$

For the lecture example 16 × 16 × 3 = 768.

Lesson: An image is a discrete signal

Scale to 0 to 1

$$x' = \frac{x}{255}$$

Turns 8 bit integers into floats.

Lesson: An image is a discrete signal

Standardise per channel

$$x'' = \frac{x' - \mu_c}{\sigma_c}$$

μ and σ are the mean and standard deviation of channel c over the training data.

Lesson: An image is a discrete signal

Grey from RGB (BT.601)

Exam favourite

$$Y = 0.299\,R + 0.587\,G + 0.114\,B$$

Green gets the biggest weight because the eye is most sensitive to it. OpenCV uses these weights.

Lesson: Colour models: RGB, CMYK, HSL and YCbCr

RGB to CMYK

$$K = 1 - \max(R', G', B'), \quad C = \frac{1 - R' - K}{1 - K}$$

$R' = R \div 255$. M and Y use G' and B' the same way.

Lesson: Colour models: RGB, CMYK, HSL and YCbCr

HSL lightness and saturation

$$L = \frac{\max + \min}{2}, \quad S = \frac{\max - \min}{1 - |2L - 1|}$$

max and min of R', G', B'. Hue is the angle on the colour wheel.

Lesson: Colour models: RGB, CMYK, HSL and YCbCr

Colour difference channels (JPEG)

$$Cb = 128 - 0.1687\,R - 0.3313\,G + 0.5\,B$$

$$Cr = 128 + 0.5\,R - 0.4187\,G - 0.0813\,B$$

A pure grey has $Cb = Cr = 128$.

Lesson: Colour models: RGB, CMYK, HSL and YCbCr

Compression ratio

Exam favourite

$$\text{ratio} = \frac{\text{original size}}{\text{compressed size}}$$

A ratio of 15 means the file is 15 times smaller.

Lesson: Why and how images are compressed

Bits per pixel

$$\text{bpp} = \frac{8 \times \text{file bytes}}{H \times W}$$

Raw colour is 24 bpp. A good JPEG photo is often 1 to 2 bpp.

Lesson: Why and how images are compressed

2D DCT of an 8 × 8 block

Exam favourite

$$F(u, v) = \tfrac{1}{4}\, C(u)\, C(v) \sum_{x=0}^{7} \sum_{y=0}^{7} f(x, y) \cos \frac{(2x + 1)u\pi}{16} \cos \frac{(2y + 1)v\pi}{16}$$

$C(0) = 1 \div \sqrt{2}$ and $C(k) = 1$ otherwise. f is the pixel value minus 128.

Lesson: The JPEG pipeline and the DCT

Quantisation

Exam favourite

$$F_q(u, v) = \text{round}\left(\frac{F(u, v)}{Q(u, v)}\right)$$

Decoding multiplies back by Q. The rounding error is the loss.

Lesson: The JPEG pipeline and the DCT

The DC coefficient

$$F(0, 0) = \frac{1}{8} \sum \sum f(x, y) = 8 \times \text{mean}$$

The top left coefficient is just 8 times the block average after subtracting 128.

Lesson: The JPEG pipeline and the DCT

Chroma subsampling 4:2:0

$$\text{data kept} = \frac{1 + \tfrac{1}{4} + \tfrac{1}{4}}{3} = \tfrac{1}{2}$$

Y keeps every pixel, Cb and Cr keep a quarter each.

Lesson: The JPEG pipeline and the DCT

Extra delay from B frames

$$\text{delay} \approx \frac{\text{longest run of B frames}}{\text{fps}} \text{ seconds}$$

Each B frame in a row makes the decoder wait one more frame time for its future reference.

Lesson: Video compression: I, P and B frames

TABLES

Table: model · channels · how it works · where it is used

Model	Channels	How it works	Where it is used
RGB	Red, green, blue	Additive: start from black and add light. All three at full give white.	Screens, cameras, most image files
CMYK	Cyan, magenta, yellow, key (black)	Subtractive: start from white paper and add ink that absorbs light.	Printing
HSL	Hue, saturation, lightness	Separates which colour, how pure, and how light.	Photo editing, colour based filtering
YCbCr	Luma Y, blue difference Cb, red difference Cr	Separates brightness from colour.	JPEG and video codecs
BGR	Blue, green, red	Same as RGB with the order reversed.	OpenCV in memory

Lesson: Colour models: RGB, CMYK, HSL and YCbCr

Table: · lossless (png) · lossy (jpeg)

	Lossless (PNG)	Lossy (JPEG)
Recovery	Exact, every pixel comes back	Approximate, some information is gone for ever
File size	Larger	Much smaller
Best for	Text, diagrams, screenshots, transparency, medical data	Photographs
Repeated saving	No change	Quality can drop each time
Typical damage	None	Blocks, blur, ringing near edges, lost fine texture

Lesson: Why and how images are compressed

Table: frame · how it is coded · strength · weakness

Frame	How it is coded	Strength	Weakness
I (intra)	On its own, like a JPEG	Can be decoded alone, good for seeking and recovering from errors	Largest
P (predicted)	From an earlier frame, using motion vectors and the leftover difference	Much smaller than I	Errors carry forward until the next I frame
B (bi predicted)	From an earlier and a later frame	Smallest	Needs a future frame, which adds delay and memory

Lesson: Video compression: I, P and B frames

TIPS AND TRAPS

COMMON MISTAKE Compression and normalisation are different things. Compression makes files smaller for storage and transfer. Normalisation changes the scale of numbers so a model can learn. The notes list this as a common confusion.

Lesson: An image is a discrete signal

COMMON MISTAKE For medical, legal and industrial images, use lossless or very light compression. The fine edges that JPEG throws away can be the tumour boundary, the digit on a number plate or the crack in a weld.

Lesson: Why and how images are compressed

2.2 Histograms, brightness and contrast

Histograms and the CDF, brightness, contrast and dynamic range, negatives, log, gamma and slicing, and histogram equalisation step by step.

KEY IDEAS

- **Histogram** h(i) counts pixels with value i, p(i) = h(i) ÷ N, CDF is the running total. It tells how many, never where.
- **Brightness, contrast, dynamic range** Brightness = mean (where the histogram sits), contrast = spread such as σ or Michelson, dynamic range = number of levels used.
- **Point operations** Negative 255 − f, linear αf + β with clipping, log c·log(1 + r) to compress highlights, gamma 255(r ÷ 255)^γ, slicing for one band.
- **Equalisation** s = round[(CDF − CDFmin) ÷ (1 − CDFmin) × 255]. Lecture example 80, 100, 120, 140 gives 0, 85, 170, 255. Not perfectly flat for discrete images.
- **Why it matters** Dark, low contrast inputs weaken edges and bias models toward dominant regions. Use the same preprocessing in training and deployment.

FORMULAS

Histogram

$$h(i) = \text{number of pixels with value } i$$

$i = 0, 1, \ldots, 255$. The bars add up to $N = w \times h$.

Lesson: The image histogram

Normalised histogram

$$p(i) = \frac{h(i)}{N}$$

The values add up to 1.

Lesson: The image histogram

Cumulative distribution

Exam favourite

$$\mathrm{CDF}(r) = \Sigma_{i \le r}\, p(i)$$

A running total that climbs from 0 to 1.

Lesson: The image histogram

Brightness

Exam favourite

$$B(I) = \frac{1}{wh}\, \Sigma_{v=1}^{h}\, \Sigma_{u=1}^{w}\, I(u, v)$$

The average intensity of all pixels.

Lesson: Brightness, contrast and dynamic range

RMS contrast

$$\sigma = \sqrt{\frac{1}{wh}\, \Sigma\, \Sigma\, (I(u, v) - B)^2}$$

The standard deviation of the intensities, the spread of the histogram.

Lesson: Brightness, contrast and dynamic range

Michelson contrast

$$C = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$$

A change in luminance relative to the average luminance, from 0 to 1.

Lesson: Brightness, contrast and dynamic range

Linear contrast and brightness

$$g = \alpha f + \beta, \text{ then clip to } [0, 255]$$

α above 1 stretches the histogram, β shifts it.

Lesson: Brightness, contrast and dynamic range

Negative

Exam favourite

$$g(x, y) = L - f(x, y)$$

L is the maximum value, 255 for 8 bits. In OpenCV: img_neg = 255 – img.

Lesson: Point operations: negative, log, gamma and slicing

Log transform

Exam favourite

$$s = c \cdot \log(1 + |r|)$$

Compresses large values more than small ones. Choose c = 255 ÷ log(256) so 255 maps to 255.

Lesson: Point operations: negative, log, gamma and slicing

Gamma (power law)

$$s = 255\, \left(\frac{r}{255}\right)^{\gamma}$$

γ below 1 brightens dark areas, γ above 1 darkens.

Lesson: Point operations: negative, log, gamma and slicing

Grey level slicing

$$s = \text{high if } A \le r \le B, \text{ otherwise low (or unchanged)}$$

Highlights one band of intensities.

Lesson: Point operations: negative, log, gamma and slicing

Equalisation mapping

Exam favourite

$$s = \mathrm{round}\left[\, \frac{\mathrm{CDF}(r) - \mathrm{CDF}_{\min}}{1 - \mathrm{CDF}_{\min}} \times 255 \,\right]$$

$\mathrm{CDF}_{\min}$ is the smallest non zero CDF value, the CDF of the darkest level present. In general replace 255 by $L - 1$.

Lesson: Histogram equalisation

TABLES

Table: photo · histogram shape · what you see

Photo	Histogram shape	What you see
Underexposed	A tall, narrow peak near the dark end	Dark, shadow detail lost
Well exposed	A wide spread across the range	Clear detail in darks and lights
Overexposed	Everything bunched near the bright end, little variation	Washed out, flat, low contrast
Low contrast (fog)	A narrow hump in the middle	Grey and dull

Lesson: The image histogram

Table: operation · use it when · example

Operation	Use it when	Example
Negative	Detail sits in dark areas and is easier to read inverted	Reading X ray films, mammograms
Linear stretch	The histogram is narrow but its shape is fine	A foggy photo
Log	A few very bright pixels hide everything else	Astronomy, the magnitude of a Fourier spectrum, satellite and medical images
Gamma	You need a smooth, adjustable brighten or darken	Display correction, under or overexposed photos
Slicing	Only one intensity band matters	Tissue in a scan, water in a satellite image, a material in inspection

Lesson: Point operations: negative, log, gamma and slicing

Table: problem in the histogram · tool · effect

Problem in the histogram	Tool	Effect
Narrow hump, good shape	Linear stretch	Spreads values, keeps their order and spacing
Narrow and unevenly crowded	Histogram equalisation or CLAHE	Spreads crowded values the most
A few huge values	Log or gamma below 1	Compresses highlights, lifts shadows
Only one band matters	Grey level slicing	Removes everything else

Lesson: Why this matters for AI models

TIPS AND TRAPS

REMEMBER Global equalisation can over boost noise in flat regions and wash out areas that were already fine. CLAHE (contrast limited adaptive histogram equalisation) equalises small tiles separately and limits how steep the curve can get. It is the version most used in medical imaging and as a preprocessing step.

Lesson: Histogram equalisation

COMMON MISTAKE These methods change the input distribution. Apply exactly the same preprocessing at training time and at deployment, or the model sees two different worlds.

Lesson: Why this matters for AI models

2.3 Noise, convolution and filtering

Noise models, convolution and output size, mean, Gaussian and median smoothing, sharpening kernels, PSNR and the convolution theorem.

KEY IDEAS

- **Noise models** Additive g = f + n (Gaussian, salt and pepper), multiplicative g = f(1 + n) (speckle). Gaussian → averaging or Gaussian filter, impulse → median, speckle → adaptive or median type, shot → denoising.
- **Convolution** Slide a kernel, multiply and add. Output size ⌊(W − k + 2P) ÷ S⌋ + 1. True convolution flips the kernel, correlation does not.
- **Smoothing** Mean (1/9)·ones, Gaussian with σ (OpenCV σ = 1.7 for 9 × 9), median for outliers. Lecture patch: mean 23.33, median 10.
- **Sharpening** Lecture high pass (1/9)[[−1, −1, −1], [−1, 8, −1], [−1, −1, −1]] sums to 0. Notebook kernel [[0, −1, 0], [−1, 5, −1], [0, −1, 0]] sums to 1 and equals identity minus Laplacian.
- **Frequency view** g = h ∗ f ⟺ G = H · F. Low pass blurs, high pass keeps edges. Noise is added on purpose for robustness tests and is the core of denoising autoencoders and diffusion models.

FORMULAS

Additive noise

Exam favourite

$$g(x, y) = f(x, y) + n(x, y)$$

f is the true image, g the one you record. The noise does not depend on the brightness.

Lesson: Four kinds of noise

Multiplicative (speckle) noise

Exam favourite

$$g(x, y) = f(x, y) \cdot (1 + n(x, y))$$

The noise is scaled by the signal, so bright regions get noisier than dark ones.

Lesson: Four kinds of noise

Filtering (correlation)

Exam favourite

$$g(x, y) = \Sigma_{i=-a}^{a} \Sigma_{j=-b}^{b} h(i, j) \cdot f(x + i, y + j)$$

A $(2a + 1) \times (2b + 1)$ kernel h. This is what cv2.filter2D and CNN layers compute.

Lesson: Convolution: sliding a kernel

True convolution

$$(h * f)(x, y) = \Sigma_i \Sigma_j h(i, j) \cdot f(x - i, y - j)$$

The same, with the kernel flipped in both directions. For symmetric kernels such as box and Gaussian the two are identical.

Lesson: Convolution: sliding a kernel

Output size

Exam favourite

$$O = \lfloor \frac{W - k + 2P}{S} \rfloor + 1$$

W input width, k kernel size, P padding on each side, S stride.

Lesson: Convolution: sliding a kernel

Mean (box) filter

Exam favourite

$$h = \frac{1}{9} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

Each output is the plain average of the 3 × 3 neighbourhood. The weights add up to 1, so flat areas keep their brightness.

Lesson: Smoothing: mean, Gaussian and median

Gaussian kernel

Exam favourite

$$G(x, y) = \frac{1}{2\pi\sigma^2} \exp(-\frac{x^2 + y^2}{2\sigma^2})$$

Near pixels count more than far ones. A common 3 × 3 version is (1/16)[[1, 2, 1], [2, 4, 2], [1, 2, 1]].

Lesson: Smoothing: mean, Gaussian and median

OpenCV default σ

$$\sigma = 0.3 \times ((k - 1) \times 0.5 - 1) + 0.8$$

Used when you pass σ = 0, as the notebook does with cv2.GaussianBlur(image, (9, 9), 0). For $k = 9$ this gives σ = 1.7.

Lesson: Smoothing: mean, Gaussian and median

Median filter

$$g(x, y) = \text{median}\{\{ f(s, t) : (s, t) \text{ in the window } \}\}$$

Not a weighted sum at all. Sort the neighbours and take the middle one.

Lesson: Smoothing: mean, Gaussian and median

Lecture high pass kernel

Exam favourite

$$h = \frac{1}{9} \begin{pmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{pmatrix}$$

The weights add up to 0. In a flat region every neighbour cancels the centre and the output is 0. At an edge the output is large.

Lesson: Sharpening and the Laplacian

Notebook sharpening kernel

Exam favourite

$$h = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

The weights add up to 1, so flat regions keep their brightness while edges get extra contrast. The project used this kernel in Stage 3.

Lesson: Sharpening and the Laplacian

Unsharp masking

$$\text{sharpened} = f + k\,(f - G_\sigma * f)$$

$f - G * f$ is the detail that a blur removes. Add k times it back.

Lesson: Sharpening and the Laplacian

Convolution theorem Exam favourite

$$g = h * f \iff G(\omega_1, \omega_2) = H(\omega_1, \omega_2)\, F(\omega_1, \omega_2)$$

Convolving in space is the same as multiplying in frequency. Capital letters are Fourier transforms.

Lesson: The frequency view and learning from noise

TABLES

Table: noise · what it looks like · typical source · best filter

Noise	What it looks like	Typical source	Best filter
Gaussian (white)	Many small random changes over the whole image, a grainy look	Sensor and electronic noise	Gaussian or averaging filter
Impulse (salt and pepper)	A few pixels jump to pure black (0) or pure white (255)	Transmission or sensor errors	Median filter
Speckle	Granular texture whose strength grows with local brightness	Ultrasound and SAR (synthetic aperture radar)	Adaptive or median type filters
Poisson (shot)	Grain that is worst in low light	Too few photons reach the sensor	Denoising with variance stabilisation

Lesson: Four kinds of noise

Table: filter · keeps · removes · looks like

Filter	Keeps	Removes	Looks like
Low pass (mean, Gaussian)	Slow changes, large shapes	Fine texture, noise, sharp edges	Blur
High pass (Laplacian, lecture 1/9 kernel)	Edges and fine detail	Flat areas and smooth shading	A dark image with bright outlines
Sharpen = original + high pass	Everything, with extra edges	Nothing	Crisper edges, more visible noise

Lesson: The frequency view and learning from noise

TIPS AND TRAPS

IN THE EXAM Salt and pepper pushes a few pixels straight to an extreme by a sudden jump. Speckle changes values gradually by multiplying them. The mechanisms differ, which is why they need different filters.

Lesson: Four kinds of noise

COMMON MISTAKE Sharpening cannot bring back detail that blur destroyed. It boosts whatever high frequencies are left, and that includes noise. In the project, sharpening the blurred COCO128 images raised mAP50 from 0.6176 to 0.6277 but it stayed below the original 0.6706. You will meet those numbers again in the project lab.

Lesson: Sharpening and the Laplacian

STEP 3 Motion and video

3.1 Optical flow, HOF, MBH and video tasks

Optical flow and the aperture problem, HOF versus MBH with every lecture scenario, the bag of visual words pipeline and modern video tasks.

KEY IDEAS

- **Optical flow** Flow (u, v) per pixel between frames T and T + 1. Magnitude √(u² + v²), direction atan2(v, u), speed = magnitude × fps. Lecture: (3, 4) gives 5 px per frame, 150 px per second at 30 fps.
- **Sparse and dense** Sparse tracks a few corners and is cheap. Dense gives every pixel a vector and suits crowded scenes but costs much more.
- **Brightness constancy** Ixu + Iyv + It = 0: one equation, two unknowns, so the aperture problem. Lucas and Kanade (window, least squares, corners) and Horn and Schunck (global smoothness).
- **HOF and MBH** HOF bins flow directions (how things move). MBH bins gradients of the flow (where motion differs) and ignores uniform camera motion. HOG is appearance.
- **Action pipeline** Interest points → patches → HOG, HOF, MBH → k-means visual words → histogram → SVM. Replaced by 3D CNNs, two stream networks and video transformers.
- **Video tasks** Interpolation predicts a missing middle frame. Also stabilisation, denoising, re-identification, propagation over time, context graphs and video RAG.

FORMULAS

Flow magnitude

Exam favourite

$$|(u, v)| = \sqrt{u^2 + v^2}$$

How far the point moved between two frames, in pixels per frame.

Lesson: Video and optical flow

Flow direction

$$\theta = \operatorname{atan2}(v, u)$$

The angle of the motion. 0° is right, 90° is up (with up taken as positive).

Lesson: Video and optical flow

Speed

Exam favourite

$$\text{speed} = \text{magnitude} \times \text{fps}$$

Pixels per frame times frames per second gives pixels per second.

Lesson: Video and optical flow

Brightness constancy

$$I(x + u, y + v, t + 1) = I(x, y, t)$$

The same point has the same brightness in the next frame, just somewhere else.

Lesson: How flow is measured: brightness constancy

Optical flow constraint

Exam favourite

$$I_x\, u + I_y\, v + I_t = 0$$

From a first order Taylor expansion of the line above. I_x, I_y are the image gradients from chapter 1.2 and I_t is the change over time.

Lesson: How flow is measured: brightness constancy

HOF bin weight

Exam favourite

$$H[b] = \Sigma_{\text{pixels with direction in bin } b} \sqrt{u^2 + v^2}$$

With 8 bins of 45°, the bin of a vector is round(θ ÷ 45°) mod 8.

Lesson: HOF and MBH: describing motion

MBH input

Exam favourite

$$\nabla u = (\partial u/\partial x,\ \partial u/\partial y), \quad \nabla v = (\partial v/\partial x,\ \partial v/\partial y)$$

If the camera adds the same vector to every pixel, every gradient stays the same. That is why MBH ignores camera motion.

Lesson: HOF and MBH: describing motion

k-means objective

Exam favourite

$$J = \Sigma_i \, \|\mathbf{x}_i - \boldsymbol{\mu}_{c(i)}\|^2$$

Alternate two steps: assign each descriptor to its nearest centre, then move each centre to the mean of its descriptors. J never increases.

Lesson: The traditional action recognition pipeline

TABLES

Table: type · what it computes · when to use it

Type	What it computes	When to use it
Sparse	Motion for a selected set of informative points, usually corners	Limited computing power, or a uniform still background where only a few objects need tracking
Dense	Motion for almost every pixel	Crowded scenes such as pedestrians in a busy street, where motion matters everywhere. It needs much more computation.

Lesson: Video and optical flow

Table: method · extra assumption · result

Method	Extra assumption	Result
Lucas and Kanade	All pixels in a small window share one (u, v). Solve the window equations together by least squares.	Fast and local. Works at corners, fails on plain edges and flat areas. Used for sparse tracking.
Horn and Schunck	The flow field is smooth over the whole image. Minimise the constraint error plus a smoothness penalty.	A dense field everywhere, but it blurs across object boundaries.

Lesson: How flow is measured: brightness constancy

Table: · hog · hof · mbh

	HOG	HOF	MBH
Question	What does it look like?	Which way is it moving?	Where does motion differ from its surroundings?
Signal	Gradient of intensity	Flow direction	Gradient of the flow
Good for	Shape of a person or object	Clear directional motion with a still camera	Camera motion, separating object from background
Weakness	Ignores motion	Can be swamped by camera motion	Less direct for a pure direction summary

Lesson: HOF and MBH: describing motion

Table: scenario · answer · why

Scenario	Answer	Why
Two people walk in opposite directions. Are their HOF features the same?	Different	HOF is direction sensitive. Their tallest bins are on opposite sides. Over the whole frame you would see two peaks.
A person walks while the camera pans right. Which descriptor isolates the person?	MBH	The pan adds uniform motion to the background, which swamps HOF. MBH suppresses uniform motion and keeps the person boundaries.
A fixed camera films a ball moving up in front of a still wall.	HOF	No competing motion. HOF shows the upward bin directly and MBH adds little.
A person cycles to the right.	Both tell different things	HOF says the dominant direction is right. MBH highlights the pedalling legs, whose motion differs from the steady torso.
A drone flies beside a running cheetah, keeping it centred. You want the leg motion.	MBH	The grass streaks past everywhere. MBH ignores that uniform motion and highlights the legs moving relative to the body.

Lesson: Which descriptor? The lecture scenarios

Table: camera · person · what the flow shows · better descriptor

Camera	Person	What the flow shows	Better descriptor
Fixed	Moving	Flow only on the person	HOF is enough
Moving	Still	Uniform flow on the background, the person moves with it in the image	MBH (and it reports almost nothing, correctly)
Moving the same way	Moving	The person can look almost still while the background streams past	MBH
Moving the opposite way	Moving	Person and background flows point opposite ways and add up in size	MBH, or HOF after removing the camera motion

Lesson: Which descriptor? The lecture scenarios

Table: hand made step · learned replacement

Hand made step	Learned replacement
Interest points, patches, HOG and HOF	3D CNNs that convolve over height, width and time together (for example C3D, I3D)
Separate appearance and motion descriptors	Two stream networks: one stream sees RGB frames (like HOG), one sees stacked optical flow (like HOF), Simonyan and Zisserman 2014
Visual word dictionary and SVM	Video transformers that attend across space and time, trained end to end

Lesson: The traditional action recognition pipeline

Table: task · what it means · notes from the lecture

Task	What it means	Notes from the lecture
Frame interpolation	Predict an unknown frame between two known frames	Used for slow motion (for example Super SloMo). Smooth interpolation in a generative model latent space shows the model learned a continuous space rather than memorising examples.
Stabilisation	Remove unwanted camera shake	A phone feature and a preprocessing step, since shake corrupts flow, HOF and MBH. Gaps created by the correction can be filled like inpainting.
Video denoising	Remove noise using several frames	Image denoising from chapter 2.3, extended over time.
Person re-identification	Find every appearance of one person across a video or cameras	Despite changes in pose, clothing colour or light. Used to follow one ID through a campus from entry gate to exit gate.
Object propagation	Share detections over time	If an object was found confidently in frame T, use that to find it cheaply in frame T + 1 instead of starting again.
Context and relations	Understand interactions, not single actions	Graph neural networks link people and objects. Example: a character interaction map and event flow built from a film.
Video question answering and RAG	Answer questions about a long video	Summarise video into text first, then retrieve the relevant part and answer from text plus video. Example: insurance claims after a car accident.

Lesson: Modern video tasks

TIPS AND TRAPS

IN THE EXAM — In flow pictures, colour shows direction and the strength of the colour shows speed. Learn to read the colour wheel, because exam figures often use it.

Lesson: Video and optical flow

REMEMBER — HOF is how things move. MBH is where motion differs. HOG is what it looks like. All three are the same operation, a local histogram of orientations, applied to three different signals.

Lesson: HOF and MBH: describing motion

IN THE EXAM The rule from the lecture: whenever camera motion is present and could contaminate HOF, prefer MBH. When nothing competes with the motion you care about, HOF alone is sufficient.

Lesson: Which descriptor? The lecture scenarios

COMMON MISTAKE Interpolation was defined in class only after several attempts. It is not generating new images and not editing pixels. It is predicting the missing frame between two frames.

Lesson: Modern video tasks

STEP 4 Recognising what is in an image

4.1 Image classification and convolutional networks

Classification datasets, CNN layers and receptive fields, SoftMax and cross entropy, confidence and calibration, and AlexNet to ViT.

KEY IDEAS

- **Classification** One label for the whole image, from a probability per class. ImageNet challenge: 1,000 classes, about 1.28 million training images. Hard because of inter-class similarity and intra-class variation.
- **CNN layer** y = ReLU(W ∗ x + b), params k × k × C_{in} × C_{out} + C_{out}, output ⌊(W − k + 2P) ÷ S⌋ + 1. Pooling shrinks maps. Features go from edges to parts to objects.
- **SoftMax and loss** $p_i = e^{z_i} \div \Sigma\, e^{z_j}$, loss $-\ln p_{true}$, gradient p − y. [2, 1, 0.1] gives [0.659, 0.242, 0.099]. Sigmoids for multi-label.
- **Confidence** Confidence is usually the SoftMax output. Inductive bias (2, 4, 6) shapes it. A calibrated model is right c of the time when it says c.
- **Architectures** AlexNet (GPU, ReLU, augmentation), VGG (stacked 3 × 3), Inception (several sizes, 1 × 1), ResNet H(x) = F(x) + x, ViT (patches as tokens).
- **Design axes** Architecture, loss, optimisation. 1 × 1 conv on 80 × 80 × 128 with 256 filters gives 80 × 80 × 256 and 33,024 parameters.

FORMULAS

One convolution layer Exam favourite

$$y = \mathrm{ReLU}(W * x + b), \quad \mathrm{ReLU}(t) = \max(0, t)$$

The filters W and biases b are learned. ReLU keeps positive responses and sets the rest to 0.

Lesson: Convolutional neural networks

Parameters of a layer Exam favourite

$$\text{params} = k \times k \times C_{\text{in}} \times C_{\text{out}} + C_{\text{out}}$$

Each of the C_{out} filters spans all C_{in} input channels and has one bias.

Lesson: Convolutional neural networks

SoftMax Exam favourite

$$p_i = \frac{e^{z_i}}{\Sigma_j\, e^{z_j}}$$

All probabilities are positive and add up to 1, so the classes compete.

Lesson: SoftMax, sigmoid and cross entropy

Sigmoid Exam favourite

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Squashes one score into (0, 1). Independent sigmoids do not add up to 1.

Lesson: SoftMax, sigmoid and cross entropy

Cross entropy loss

Exam favourite

$$L = -\Sigma_j \, y_j \ln p_j = -\ln p_{\text{true}}$$

y is one hot: 1 for the true class, 0 elsewhere. A confident correct answer gives a loss near 0.

Lesson: SoftMax, sigmoid and cross entropy

Gradient

$$\partial L \div \partial z_k = p_k - y_k$$

Push the true logit up and the others down, each in proportion to its error.

Lesson: SoftMax, sigmoid and cross entropy

Calibration

Exam favourite

Among all predictions made with confidence c, the fraction that is correct should be about c

A model that says 0.80 but is right only half the time is overconfident.

Lesson: Confidence, inductive bias and calibration

Residual learning

Exam favourite

$$H(x) = F(x) + x$$

The shortcut passes x straight through. The layers only learn the difference $F(x)$.

Lesson: From AlexNet to Vision Transformers

ViT tokens

$$\text{number of tokens} = \left(\frac{H}{P}\right) \times \left(\frac{W}{P}\right)$$

For a 224 × 224 image and 16 × 16 patches: 14 × 14 = 196 tokens.

Lesson: From AlexNet to Vision Transformers

Gradient descent

Exam favourite

$$w \leftarrow w - \eta \, \frac{\partial L}{\partial w}$$

η is the learning rate. Too large and training jumps around, too small and it crawls.

Lesson: Designing a model: architecture, loss, optimisation

TABLES

Table: class · predicted probability

Class	Predicted probability
Dog	0.82
Cat	0.10
Wolf	0.05
Other	0.03

Lesson: Image classification and its datasets

Table: difficulty · meaning · example

Difficulty	Meaning	Example
Many classes	1,000 categories, many of them close	Over a hundred dog breeds
Inter-class similarity	Different classes can look alike	Wolf and husky, truck and bus
Intra-class variation	One class can look very different	A chair from the side, from above, folded, in the dark

Lesson: Image classification and its datasets

Table: · underfitting · overfitting

	Underfitting	Overfitting
Cause	Model too simple for the data	Model memorises the training examples and their noise
Training score	Poor	High
Test score	Poor	Poor
Typical fix	Bigger model, more training	More data, augmentation, regularisation, early stopping

Lesson: Image classification and its datasets

Table: model · year · key idea

Model	Year	Key idea
AlexNet	2012	Deep CNN trained on GPUs with ReLU, dropout regularisation and data augmentation. A large jump over hand made features.
VGG	2014	Only small 3 × 3 filters, stacked deep. Repeated small filters reach as far as large ones with fewer weights and more non linearity.
GoogLeNet (Inception)	2014	Filters of several sizes side by side, because objects contain patterns of different sizes. 1 × 1 convolutions keep it cheap.
ResNet	2015	Shortcut connections, so each block learns a correction F(x) = H(x) − x. Made networks of 100 and more layers trainable.
Vision Transformer (ViT)	2020	Cut the image into patches and treat them as a sequence of tokens. Attention links distant parts directly.

Lesson: From AlexNet to Vision Transformers

Table: axis · question it answers · examples

Axis	Question it answers	Examples
Architecture	Which functions can the model represent?	Convolutions, residual blocks, attention, feature pyramids
Loss	What do we ask it to do?	Cross entropy, Huber, focal loss (chapter 5.2)
Optimisation	How do we search for good weights?	SGD, momentum, Adam, learning rate schedules

Lesson: Designing a model: architecture, loss, optimisation

TIPS AND TRAPS

IN THE EXAM Classification gives one label for the whole image. Detection adds a box for each of a variable number of objects. Segmentation labels every pixel. Know all three outputs.

Lesson: Image classification and its datasets

REMEMBER Computers overflow when they raise e to a large number. Since SoftMax(z) = SoftMax(z − max z), real code subtracts the largest logit first. The answer does not change.

Lesson: SoftMax, sigmoid and cross entropy

COMMON MISTAKE Some models now report lower error than the estimated human error on ImageNet. That does not mean human level vision. Models still fail under unusual conditions, distribution shift, adversarial changes, occlusion, new categories and ambiguous scenes. Benchmark wins are specific to the benchmark.

Lesson: From AlexNet to Vision Transformers

STEP 5 Finding objects

5.1 From sliding windows to anchor boxes

Detector outputs, sliding window cost, anchors with box encoding and decoding, anchor grids and heads, and trusting confidence scores.

KEY IDEAS

- **Detector output** For each object: class, box, confidence. Corners (x1, y1, x2, y2) or centre size (xc, yc, w, h). The number of objects varies.
- **Sliding windows** (⌊(W − w) ÷ s⌋ + 1)(⌊(H − h) ÷ s⌋ + 1) windows per shape: 3,285 for 64 × 128 at stride 8 in 640 × 480. Four problems: location, scale, shape, computation.
- **Anchors and regression** Rough box plus four learned corrections. t = ((x − xa) ÷ wa, (y − ya) ÷ ha, ln(w ÷ wa), ln(h ÷ ha)), decode with x = xa + tx wa and w = wa e^tw.

- **Heads and anchor design** One backbone pass, conv heads score all anchors. k(C + 1) class and 4k box channels. 9 anchors = 3 scales × 3 ratios. SSD 8,732 boxes. Stride s ≤ 0.367w.
- **Trusting scores** Aleatoric (data noise) versus epistemic (model ignorance). SoftMax is valid but often overconfident. ECE measures it. Conformal prediction gives a coverage guarantee.

FORMULAS

Two box formats

Exam favourite

$$(x_1, y_1, x_2, y_2) \longleftrightarrow (x_c, y_c, w, h)$$

$x_c = (x_1 + x_2) \div 2$, $w = x_2 - x_1$, and the same for y. Image y grows downwards.

Lesson: What a detector predicts

Detection output

$$\{\{\text{ class, box, confidence }\}\} \times N \text{ objects}$$

N is not known in advance. It can be 0 or 50.

Lesson: What a detector predicts

Number of window positions

Exam favourite

$$N = (\lfloor \frac{W - w}{s} \rfloor + 1) \times (\lfloor \frac{H - h}{s} \rfloor + 1)$$

Image $W \times H$, window $w \times h$, stride s. For every window shape and scale.

Lesson: Sliding windows and their four problems

Encode (make the training target)

Exam favourite

$$t_x = \frac{x - x_a}{w_a}, \quad t_y = \frac{y - y_a}{h_a}, \quad t_w = \ln \frac{w}{w_a}, \quad t_h = \ln \frac{h}{h_a}$$

Anchor (x_a, y_a, w_a, h_a), true box (x, y, w, h), all in centre size form.

Lesson: Anchor boxes and box regression

Decode (turn a prediction into a box)

Exam favourite

$$x = x_a + t_x w_a, \quad y = y_a + t_y h_a, \quad w = w_a e^{t_w}, \quad h = h_a e^{t_h}$$

The exact reverse of encoding.

Lesson: Anchor boxes and box regression

Heads as convolutions

$$\text{class scores} = \text{SoftMax}(\text{Conv}(V, W_{cls})), \quad \text{offsets} = \text{Conv}(V, W_{loc})$$

V is the feature map. Two heads, one backbone.

Lesson: Anchor grids, shapes and convolutional heads

Head sizes

Exam favourite

$$\text{class channels} = k(C + 1), \quad \text{box channels} = 4k$$

k anchors per position, C classes plus background.

Lesson: Anchor grids, shapes and convolutional heads

Anchor stride rule

$$s \le 0.367\, w$$

Worst case: the object sits midway between four grid points. Requiring IoU ≥ 0.5 there gives $(w - s/2)^2 \ge \frac{2}{3} w^2$, which simplifies to this.

Lesson: Anchor grids, shapes and convolutional heads

TABLES

Table: task · answers · output

Task	Answers	Output
Classification	What is in the image?	One class label
Localisation	What and where, one object	One label and one box
Detection	What and where, many objects	A list of (class, box, score) of any length
Segmentation	Where, pixel by pixel	A class for every pixel

Lesson: What a detector predicts

Table: application · which mistake hurts more

Application	Which mistake hurts more
Self driving and driver assistance	A missed pedestrian (false negative), so favour recall
Surveillance	A missed threat, but too many false alarms cause alarm fatigue
Crop disease spotting	Missed disease loses the crop, false alarms waste spray
Medical screening	A missed lesion, so favour recall and let a doctor confirm

Lesson: What a detector predicts

Table: problem · what goes wrong

Problem	What goes wrong
Location	A small step is accurate but slow. A large step is fast but can cut an object in half and miss it.
Scale	Objects can be tiny or huge. A small window loses context, a fixed window cannot fit both a far person and a near bus.
Shape	People are tall, cars are wide. One square window fits neither.
Computation	Thousands of windows, each a full classifier pass, recomputing almost the same pixels as its neighbours.

Lesson: Sliding windows and their four problems

Table: kind · source · does more data help?

Kind	Source	Does more data help?
Aleatoric	Noise in the data itself: blur, occlusion, a tiny object	No
Epistemic	The model does not know: too little or unrepresentative data, wrong assumptions	Yes

Lesson: How far can we trust the score?

TIPS AND TRAPS

IN THE EXAM — Overlap between windows is a consequence, not a separate problem. A fine stride makes neighbouring windows overlap heavily, and that is why the same pixels are processed again and again.

Lesson: Sliding windows and their four problems

REMEMBER — This is why anchor sizes grow with the stride of each feature level. RetinaNet uses anchors of 32 on its stride 8 level: 8 ÷ 32 = 0.25, safely under 0.367.

Lesson: Anchor grids, shapes and convolutional heads

5.2 Training a detector: targets and losses

Anchor targets by IoU, Huber loss for boxes, class imbalance, cross entropy and focal loss, bias initialisation and architecture versus loss.

KEY IDEAS

- **Target assignment** Match anchors to true boxes by IoU. RetinaNet: ≥ 0.5 positive, < 0.4 background, between ignored. RPN: 0.7 and 0.3. Background gets only a class loss.
- **Box loss** MSE squares errors (outliers dominate), MAE is robust but kinked, Huber = ½e² for |e| ≤ 1 else |e| − ½, gradient clip(e, −1, 1). Bad box example: 345× under e², 1.1× in gradient with Huber.
- **Imbalance** Easy background swamps the loss: 97.8% in the worked example. Remedies: subsample, down-weight, hard example mining, focal loss.
- **Cross entropy and focal loss** BCE = $-\ln p_t$. FL = $-\alpha(1 - p_t)^{\gamma} \ln p_t$, $\gamma = 2$, $\alpha = 0.25$. Background share falls to 0.54%. Bias $b = \ln(\pi \div (1 - \pi)) = -4.595$ for $\pi = 0.01$.

- **Design axes** Architecture, loss, optimisation. GAN → DCGAN (architecture) → WGAN (loss, earth mover distance).

FORMULAS

Per-anchor loss

Exam favourite

$$L = \frac{1}{N} \Sigma_{\text{anchors}} [\, \alpha L_{\text{cls}} + \beta\, \delta(\text{positive})\, L_{\text{loc}} \,]$$

δ is 1 for a positive anchor and 0 otherwise, so background anchors only pay the class loss. *N* is often the number of positive anchors.

Lesson: Giving every anchor a target

Mean squared error

$$\text{MSE} = \frac{1}{n} \Sigma\, (\hat{y}_i - y_i)^2$$

Squares the error, so large errors are punished far more: $0.1^2 = 0.01$ but $10^2 = 100$.

Lesson: Box loss: MSE, MAE and Huber

Mean absolute error

$$\text{MAE} = \frac{1}{n} \Sigma\, |\hat{y}_i - y_i|$$

Grows linearly, so outliers matter less. Its best constant guess is the median, MSE's is the mean.

Lesson: Box loss: MSE, MAE and Huber

Huber (Smooth L1)

Exam favourite

$$\text{Huber}(e) = \tfrac{1}{2}e^2 \text{ if } |e| \le 1, \quad |e| - \tfrac{1}{2} \text{ otherwise}$$

e is prediction minus target, not Euler's number. The general form uses a threshold δ.

Lesson: Box loss: MSE, MAE and Huber

Huber slope

$$\partial\text{Huber} \div \partial e = \text{clip}(e, -1, 1)$$

Proportional for small errors, capped for large ones.

Lesson: Box loss: MSE, MAE and Huber

Entropy

$$H(p) = -\Sigma_k\, p_k \log p_k$$

Average surprise. Highest when all outcomes are equally likely, zero when the outcome is certain. The lecturer's picture: a messy table has high entropy, a tidy one low.

Lesson: Class imbalance and cross entropy

Binary cross entropy

Exam favourite

$$\text{BCE} = -[\, y \ln p + (1 - y) \ln(1 - p) \,]$$

If $y = 1$ only the first term is active, if $y = 0$ only the second.

Lesson: Class imbalance and cross entropy

p_t notation

$$p_t = p \text{ if } y = 1, \quad 1 - p \text{ if } y = 0, \quad \text{so} \quad \text{BCE} = -\ln p_t$$

p_t is the probability given to the correct answer.

Lesson: Class imbalance and cross entropy

Focal loss

Exam favourite

$$\text{FL}(p_t) = -\alpha_t\, (1 - p_t)^{\gamma} \ln p_t$$

The factor $(1 - p_t)^{\gamma}$ goes to 0 for easy, confident examples and stays near 1 for hard ones. $\gamma = 0$ gives cross entropy. RetinaNet used $\gamma = 2$ and $\alpha = 0.25$ for the foreground class.

Lesson: Focal loss and the RetinaNet bias trick

Prior bias

Exam favourite

$$\sigma(b) = \pi \implies b = \ln \frac{\pi}{1 - \pi}$$

With $\pi = 0.01$: $b = -\ln 99 \approx -4.595$. Every anchor begins by saying probably background.

Lesson: Focal loss and the RetinaNet bias trick

Earth mover distance in 1D

$$W_1 = \Sigma_i\, |F_P(i) - F_Q(i)|$$

For unit spacing, using the cumulative sums *F*. It is the least total work to move one pile of earth into the shape of the other.

Lesson: Architecture or loss: the GAN story

TABLES

Table: detector · positive · background

Detector	Positive	Background
RetinaNet	IoU ≥ 0.5	IoU < 0.4
Faster R-CNN region proposals	IoU > 0.7 (or the best anchor for an object)	IoU < 0.3

Lesson: Giving every anchor a target

Table: remedy · what it does · example

Remedy	What it does	Example
Subsample negatives	Use only some background anchors	SSD keeps 3 negatives per positive, the Faster R-CNN RPN samples 256 anchors per image
Down-weight negatives	Keep all, but give background a smaller weight	α-balanced cross entropy
Online hard example mining	Compute every loss, keep only the hardest negatives	OHEM (Shrivastava and colleagues, 2016)
Focal loss	Reshape the loss so easy examples contribute almost nothing	RetinaNet (Lin and colleagues, 2017)

Lesson: Class imbalance and cross entropy

Table: p t · ce · weight γ = 2 · fl (γ = 2)

p_t	CE	weight γ = 2	FL (γ = 2)
0.1 (hard)	2.3026	0.81	1.8651
0.5	0.6931	0.25	0.1733
0.9 (easy)	0.1054	0.01	0.0011
0.99 (very easy)	0.0101	0.0001	0.000001

Lesson: Focal loss and the RetinaNet bias trick

Table: step · what changed · axis

Step	What changed	Axis
GAN (2014)	The discriminator is a binary classifier: real or fake, sigmoid and cross entropy	Original
DCGAN (2015)	Convolutional layers, batch normalisation, no fully connected layers, which made GANs train stably	Architecture
WGAN (2017)	The loss becomes the Wasserstein (earth mover) distance. The critic outputs a real number, not a probability	Loss

Lesson: Architecture or loss: the GAN story

5.3 IoU, NMS and detector architectures

IoU, NMS and Soft-NMS, one stage and two stage detectors, RetinaNet with FPN, Faster R-CNN, bilinear interpolation and RoIAlign.

KEY IDEAS

- **IoU** $|A \cap B| \div (|A| + |B| - |A \cap B|)$. Lecture: 0.389 (miss) and 0.747 (hit) at 0.5. Equal squares shifted by s: $(1 - s) \div (1 + s)$.
- **NMS** Sort by score, keep the best, remove same class boxes with IoU above the threshold, repeat. Slide example gives {A, D}. Crowds and fixed thresholds are weak spots. Soft-NMS decays by $e^{-IoU^2 \div \sigma}$.

- **One and two stage** One stage: SSD, YOLO, RetinaNet. Two stage: Faster R-CNN with a class agnostic RPN. The speed and accuracy divide is fuzzy today. Stage wise, joint and hybrid training.
- **RetinaNet and FPN** ResNet + FPN + focal loss + prior bias. Early layers: high resolution, low meaning. FPN: 1 × 1 lateral + ×2 upsample + add, shared heads. 76,725 anchors at 640 × 640.
- **RoIAlign** Bilinear f = Σ weights × corners, weights are opposite areas and add to 1. (2.3, 4.6) gives 26.8. RoIAlign samples bins without rounding, so the detector trains end to end.

FORMULAS

IoU

Exam favourite

$$\mathrm{IoU}(A, B) = \frac{|A \cap B|}{|A \cup B|} = \frac{|A \cap B|}{|A| + |B| - |A \cap B|}$$

Between 0 and 1. The overlap rectangle runs from the larger of the left edges to the smaller of the right edges, and the same for top and bottom.

Lesson: Intersection over union

Soft-NMS

$$s_i \leftarrow s_i \cdot e^{-\mathrm{IoU}(M,\, b_i)^2 \div \sigma}$$

Instead of deleting, decay the score of each box by how much it overlaps the kept box *M*. Near duplicates sink, slightly overlapping neighbours survive.

Lesson: Removing duplicates with NMS

One FPN merge

Exam favourite

$$P_l = \mathrm{Conv}_{1\times1}(C_l) + \mathrm{Upsample}_{\times2}(P_{l+1})$$

The 1 × 1 convolution matches the channel count (to 256), the upsampling matches the size, then the two maps are added. A 3 × 3 convolution then smooths each result.

Lesson: RetinaNet and the feature pyramid

Anchors scored

$$\Sigma_{\text{levels}}\ (\text{cells}) \times A$$

For 640 × 640: (6,400 + 1,600 + 400 + 100 + 25) × 9 = 76,725 anchors.

Lesson: RetinaNet and the feature pyramid

Bilinear interpolation

Exam favourite

$$f(x, y) = (1 - u)(1 - v)\, f_{00} + u(1 - v)\, f_{10} + (1 - u)v\, f_{01} + uv\, f_{11}$$

u and *v* are the fractional parts of *x* and *y*. Each corner's weight is the area of the rectangle opposite it. The weights add to 1.

Lesson: Faster R-CNN, interpolation and RoIAlign

TABLES

Table: problem · what happens

Problem	What happens
Crowds	Two real people who overlap by more than the threshold: the lower scoring one is deleted.
Per class only	A dog box and a cat box on the same animal both survive.
One fixed threshold	Too low removes real neighbours (lower recall), too high keeps duplicates (lower precision).

Lesson: Removing duplicates with NMS

Table: · one stage · two stage

	One stage	Two stage
Idea	Image straight to boxes and classes	Propose likely regions first, then refine each one
Examples	SSD, YOLO, RetinaNet	R-CNN, Fast R-CNN, Faster R-CNN, Mask R-CNN
Structure	Backbone with a class head and a box head	Backbone, region proposals, crop each region, then classify and refine
Strength	Simple operations, easy to run fast	Per object features, easy to add masks or keypoints

Lesson: One stage or two stage

Table: mode · how · assumes

Mode	How	Assumes
Stage wise	Train stage 1 with its loss, feed its output to stage 2 and train that separately	The stages are independent
Joint (end to end)	Add all the losses and back-propagate from the last stage to the first	The stages depend on each other
Hybrid	Stage wise first, then joint fine tuning, or freeze stage 1 and train stage 2	Stability first, then end to end tuning

Lesson: One stage or two stage

Table: · early layers · deep layers

	Early layers	Deep layers
Spatial resolution	High, for example 80 × 80	Low, for example 20 × 20
Semantic meaning	Low: edges, colours, blobs	High: parts and whole objects
Receptive field	Small	Large, sees context

Lesson: RetinaNet and the feature pyramid

TIPS AND TRAPS

COMMON MISTAKE The lecturer mentioned comparing with the ground truth. At test time there is no ground truth, so NMS uses only scores and box to box IoU. Ground truth is used in training (target assignment) and in evaluation.

Lesson: Removing duplicates with NMS

5.4 Evaluating and choosing detectors

Datasets and the long tail, TP and FP matching, precision and recall, AP, mAP and COCO AP, the Pareto frontier, batch norm with small batches and transfer learning.

KEY IDEAS

- **Matching** Sort by score. TP if IoU above α with an unmatched true box, otherwise FP (poor box, background or duplicate). Unmatched true boxes are FN. No true negatives. Three dogs example: TP 2, FP 2, FN 1.
- **Precision and recall** P = TP ÷ (TP + FP), R = TP ÷ (TP + FN), F1 = 2PR ÷ (P + R). Example 0.5, 0.667, 0.571. A higher score threshold usually raises P and lowers R.
- **AP** Envelope $p_{interp}(r)$ = max precision at recall ≥ r. All point AP adds rectangles where recall grows. Lecture list gives 0.674, and the 11 point version also 0.674. COCO uses 101 points.
- **mAP and COCO** mAP = mean AP over classes. COCO AP also averages IoU 0.50 to 0.95, so 0.80 at IoU 0.5 became 0.559. AP_S below 32^2, AP_L above 96^2.
- **Practice** Dominated models are never the right pick. Lower resolution and fewer proposals speed things up. Batch 32 to 2 makes BN 4 times noisier, so use frozen BN, GroupNorm or Sync-BN. Pre-training helps: 34.3 to 36.7 to 37.4. Long tail: r = max(1, √(t ÷ f)).

FORMULAS

Repeat factor sampling (LVIS)

Exam favourite

$$r(c) = \max(1, \sqrt{t \div f(c)}), \quad r(\text{image}) = \max \text{ over its classes } r(c)$$

$f(c)$ is the fraction of training images that contain class c and t is a threshold, 0.001 in the LVIS paper. Images with rare classes are shown r times per epoch.

Lesson: Datasets and the long tail

Precision

Exam favourite

$$\mathrm{P} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}$$

Of the boxes the model drew, how many were right. False alarms lower it.

Lesson: Precision, recall and the score threshold

Recall

Exam favourite

$$\mathrm{R} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}$$

Of the real objects, how many were found. Misses lower it.

Lesson: Precision, recall and the score threshold

F1 score

$$\mathrm{F}_1 = \frac{2\mathrm{PR}}{\mathrm{P} + \mathrm{R}}$$

The harmonic mean. It is low if either one is low.

Lesson: Precision, recall and the score threshold

Interpolated precision

Exam favourite

$$p_{\text{interp}}(r) = \max_{r' \geq r} p(r')$$

The best precision at this recall or any higher recall. It turns the zig-zag into a staircase that only goes down.

Lesson: The PR curve and average precision

AP, all points

Exam favourite

$$\mathrm{AP} = \Sigma_k \, (r_k - r_{k-1}) \cdot p_{\text{interp}}(r_k)$$

Add a rectangle each time recall increases.

Lesson: The PR curve and average precision

AP, 11 points

$$\mathrm{AP}_{11} = \frac{1}{11} \, \Sigma_{r \in \{0,\, 0.1,\, \ldots,\, 1\}} \, p_{\text{interp}}(r)$$

PASCAL VOC 2007. COCO uses 101 points, 0 to 1 in steps of 0.01.

Lesson: The PR curve and average precision

mAP

Exam favourite

$$\mathrm{mAP} = \frac{1}{C} \, \Sigma_c \, \mathrm{AP}_c$$

The mean over the C classes.

Lesson: mAP and the COCO protocol

COCO AP

Exam favourite

$$\mathrm{AP}_{\mathrm{COCO}} = \frac{1}{10} \, \Sigma_{t \in \{0.50,\, 0.55,\, \ldots,\, 0.95\}} \, \mathrm{mAP}_t$$

Mean over classes and over IoU thresholds.

Lesson: mAP and the COCO protocol

Noise of a batch statistic

$$\mathrm{SE}(\hat{\mu_{\mathrm{B}}}) = \frac{\sigma}{\sqrt{N}}$$

σ is the spread of the activation and N the batch size.

Lesson: Choosing, speeding up and training a detector

TABLES

Table: dataset · classes · images (slide) · notes

Dataset	Classes	Images (slide)	Notes
PASCAL VOC	20	about 5K	The classic small benchmark. The 2007 train and validation split has 5,011 images.
MS COCO	80	about 120K	The standard. The lecturer: any good paper will use this.
Open Images	600	1.7M	Very large, many boxes per image.
LVIS	about 1,000 on the slide	50K	Large vocabulary with an extreme long tail. The released version 1.0 has 1,203 categories.

Lesson: Datasets and the long tail

Table: imbalance · where · typical cure

Imbalance	Where	Typical cure
Foreground against background	Inside every image, most anchors see background	Focal loss, hard example mining (chapter 5.2)
Long tail	Across the dataset, some classes are rare	Repeat factor sampling, class balanced loss weights, decoupled training

Lesson: Datasets and the long tail

Table: threshold · meaning · suits

Threshold	Meaning	Suits
$\alpha = 0.5$	Roughly the right place. The PASCAL VOC rule.	Counting people in a crowd
$\alpha = 0.75$	A tight box	A robot grasping an object, measuring size
0.5 to 0.95 averaged	Rewards tight boxes at every level. The COCO rule.	Comparing detectors fairly

Lesson: Matching detections: TP, FP and FN

Table: outcome · rule

Outcome	Rule
True positive (TP)	Right class, IoU above α with a true box that has not been matched yet
False positive (FP)	IoU below α (poor box), lies on background, or a duplicate of an object already matched
False negative (FN)	A true object that no detection matched. Also called a miss.
True negative	Not used. The number of empty windows is effectively infinite.

Lesson: Matching detections: TP, FP and FN

Table: threshold · kept · tp · fp

Threshold	Kept	TP	FP	Precision	Recall
≥ 0	6	3	3	0.50	1.00
≥ 0.05	5	3	2	0.60	1.00
≥ 0.2	4	3	1	0.75	1.00
≥ 0.4	3	2	1	0.67	0.67
≥ 0.6	2	2	0	1.00	0.67
≥ 0.95	1	1	0	1.00	0.33

Lesson: Precision, recall and the score threshold

Table: detector on coco (slide 92) · map

Detector on COCO (slide 92)	mAP
SSD with MobileNet, low resolution	21
SSD with Inception V2, low resolution	24
R-FCN with ResNet-101, 300 proposals	30
Faster R-CNN with ResNet-101, 100 proposals	32
Faster R-CNN with Inception-ResNet-V2, 300 proposals	37

Lesson: mAP and the COCO protocol

Table: data · resolution · pixels · compared with 224 × 224

Data	Resolution	Pixels	Compared with 224 × 224
MNIST	28 × 28	784	0.02×
ImageNet	224 × 224	50,176	1×
COCO	640 × 640	409,600	8.2×
COCO	800 × 1333	1,066,400	21.3×

Lesson: Choosing, speeding up and training a detector

TIPS AND TRAPS

REMEMBER Always look at the class histogram of a dataset before comparing mAP numbers. A model can look good on average while failing every rare class.

Lesson: Datasets and the long tail

IN THE EXAM Remember the names this way. The first word says whether the model was right, the second says what it predicted. A false positive is a box the model drew that was wrong. The lecturer added that books draw the confusion matrix in different layouts, so learn the idea and not one picture.

Lesson: Matching detections: TP, FP and FN

COMMON MISTAKE Check which metric a paper reports. COCO AP numbers of 30 to 50 look small next to PASCAL VOC AP@0.5 numbers of 70 to 90, but they measure a much stricter thing.

Lesson: mAP and the COCO protocol

STEP 6 Understanding every pixel

6.1 What segmentation predicts

Pixel level classification, semantic, instance and panoptic masks, SoftMax per pixel, receptive field, the downsampling trade-off and dense prediction.

KEY IDEAS

- **The task** $I \in \mathbb{R}^{H\times W\times 3} \to Y \in \{1..C\}^{H\times W}$. Before argmax H × W × C, after argmax H × W. 512 × 512 needs 262,144 predictions. Use the cheapest task that answers the question.
- **Semantic, instance, panoptic** Semantic: class per pixel, cannot count touching objects. Instance: object identity. Panoptic: both, scored by PQ = SQ × RQ.
- **SoftMax per pixel** $p_c = e^{z_c} \div \Sigma\, e^{z_k}$. argmax of SoftMax = argmax of logits. SoftMax for one label per pixel, sigmoids for multi-label. Ignore label 255 gets no loss.
- **Why classifiers fail** Convolution is equivariant, global pooling is invariant. Downsampling grows the receptive field (r += (k − 1)dj) but erases thin and small structures. What the encoder drops, the decoder cannot recover.
- **WHAT and WHERE** Early layers: high resolution, WHERE. Deep layers: semantics, WHAT. Every architecture in the next chapter combines the two. Same body, different head for depth, flow and keypoints.

FORMULAS

Semantic segmentation

Exam favourite

$$I \in \mathbb{R}^{H \times W \times 3} \rightarrow Y \in \{1, \ldots, C\}^{H \times W}$$

One classification repeated at every pixel. Tasks that predict something at every pixel are called **dense prediction**.

Lesson: From one label to a label per pixel

Panoptic quality

$$\mathrm{PQ} = \frac{\Sigma_{\text{matched}}\ \mathrm{IoU}}{|\mathrm{TP}| + \tfrac{1}{2}|\mathrm{FP}| + \tfrac{1}{2}|\mathrm{FN}|} = \mathrm{SQ} \times \mathrm{RQ}$$

Segmentation quality (mean IoU of matched segments) times recognition quality (an F1 style count). Three matches with IoU 0.8, 0.7, 0.9, one FP and one FN: SQ = 0.80, RQ = 3 ÷ 4 = 0.75, PQ = 0.60.

Lesson: From one label to a label per pixel

SoftMax at each pixel

Exam favourite

$$P(c \mid x, y) = \frac{e^{z_c(x,\,y)}}{\Sigma_k\, e^{z_k(x,\,y)}}$$

Positive, and the C values add to 1. Road competes with the other four classes, car with the other four, and so on.

Lesson: What the network outputs

Label

$$Y(x, y) = \operatorname{argmax}_c\, P(c \mid x, y)$$

Before argmax: H × W × C. After argmax: H × W.

Lesson: What the network outputs

Receptive field recurrence

Exam favourite

$$r_l = r_{l-1} + (k - 1) \cdot d_l \cdot j_{l-1}, \quad j_l = j_{l-1} \cdot s_l$$

Start with r = 1 and j = 1. k is the kernel size, d the dilation, s the stride, j the jump between neighbouring cells in input pixels.

Lesson: Why a classifier loses the WHERE

TABLES

Table: task · question · output · size

Task	Question	Output	Size
Classification	What is this image?	One label	1 label (C scores)
Detection	What, and roughly where?	Class, box and score per object	N × (4 + 1 + C)
Semantic segmentation	What, and exactly where?	A class for every pixel (a mask)	H × W labels (H × W × C scores)

Lesson: From one label to a label per pixel

Table: field · classes · why shape matters

Field	Classes	Why shape matters
Autonomous driving	road, sidewalk, vehicles, pedestrians	Free space is a region, not a box
Medical imaging	tumour, organ, tissue	Area, margin, surgical and radiation planning
Satellite images	water, vegetation, road, building	How much land belongs to each class
Agriculture	crop, weed, soil	Laser weeding must not burn the crop next to the weed

Lesson: From one label to a label per pixel

Table: · semantic · instance · panoptic

	Semantic	Instance	Panoptic
Answers	Which class is each pixel?	Which object is each pixel?	Both, for every pixel
Three people	All labelled person	Person 1, 2 and 3	People separate, sky and sand labelled too
Counting	Not possible from the mask	Yes	Yes, for countable things
Typical model and metric	FCN, U-Net, DeepLab, mIoU	Mask R-CNN, mask AP	Panoptic FPN, Mask2Former, PQ

Lesson: From one label to a label per pixel

Table: classification wants · segmentation needs

Classification wants	Segmentation needs
Invariance: location does not matter	Precision: every boundary matters
A summary of the whole image	Small objects must not disappear
One vector, one label	The output must keep the spatial grid

Lesson: Why a classifier loses the WHERE

Table: downsampling improves · downsampling loses

Downsampling improves	Downsampling loses
Larger receptive field	Fine boundaries
More abstract, semantic features	Thin structures such as lane markings and blood vessels
Context, and less computation	Small objects

Lesson: Why a classifier loses the WHERE

Table: · early layers · deep layers

	Early layers	Deep layers
Resolution	High	Low
Good at	Edges, textures, boundaries (WHERE)	Objects and context (WHAT)
Weakness	Weak meaning	Coarse location

Lesson: Why a classifier loses the WHERE

Table: task · output · value · typical loss

Task	Output	Value	Typical loss
Classification	1 prediction (C scores)	category	cross-entropy
Detection	N × (class, box, score)	category and 4 numbers	CE or focal, plus Smooth-L1
Semantic segmentation	H × W × C	category per pixel	pixel CE, Dice, focal
Depth estimation	H × W × 1	distance	L1, scale invariant log loss
Optical flow	H × W × 2	motion (u, v) per pixel	end point error
Surface normals	H × W × 3	unit vector	angular or cosine loss
Keypoint heat maps	H × W × K	probability per joint	MSE on Gaussian blobs

Lesson: The dense prediction family

TIPS AND TRAPS

REMEMBER Dense labels are expensive. Someone must trace every boundary. When masks are scarce, people use weak labels (tags, boxes, points or scribbles), a few masks plus many unlabelled images, self supervised pre-training, or a click based tool such as SAM to speed up annotation.

Lesson: What the network outputs

IN THE EXAM The slide warned: do not memorise architecture names. FCN, U-Net, DeepLab and transformers are different answers to one question: how do we combine WHAT is present with exactly WHERE it is?

Lesson: Why a classifier loses the WHERE

REMEMBER Same body, different head. An FCN, U-Net or DeepLab body is reused across the family. Change the last layer from C class channels to 1 channel (depth), 2 (flow) or K (heat maps) and change the loss to match.

Lesson: The dense prediction family

6.2 Segmentation architectures

FCN, upsampling and transposed convolution, encoder-decoders and skip connections, U-Net, dilated convolution, ASPP and DeepLab.

KEY IDEAS

- **FCN** An FC layer equals a conv with the map size as kernel, so classifiers become dense predictors. FCN-32s, 16s and 8s predict at 1/32, 1/16 and 1/8 and fuse earlier maps.
- **Upsampling** Nearest copies, bilinear blends (0, 25, 75, 100 for the lecture map), transposed conv learns. $H_{out} = (H_{in} - 1)s - 2p + k$. Use k = 2s to avoid checkerboards. Masks: nearest only.
- **Encoder-decoder and skips** Encoder: what. Decoder: where. Bottleneck: strongest context, coarsest detail. Skips carry WHERE. Concat costs about twice addition (73,792 against 36,928).
- **U-Net** Contracting path, bottleneck, expanding path, skip at every level. Original 572 → 388 with valid convs, 31.0 M parameters, about half in the bottleneck.
- **DeepLab** Dilated 3 × 3: $k_{eff} = 2r + 1$, 9 weights, padding r. Rates 1, 2, 4, 8 give a field of 31. ASPP: rates 6, 12, 18 + 1 × 1 + pooling. v3+ adds a decoder at stride 4.

FORMULAS

Output size Exam favourite

$$H_{out} = (H_{in} - 1) \cdot s - 2p + k + p_{out}$$

s stride, p padding, k kernel, p_{out} extra output padding. With $H_{in} = 16$, k = 4, s = 2, p = 1 the size doubles to 32.

Lesson: Three ways to make a map bigger

Dilated 3 × 3 kernel Exam favourite

$$k_{eff} = k + (k - 1)(r - 1) = 2r + 1$$

r = 1, 2, 4, 6, 12, 18 cover 3, 5, 9, 13, 25 and 37 cells. Always 9 weights. Padding r keeps the map size.

Lesson: Dilated convolution, ASPP and DeepLab

TABLES

Table: variant · uses · map for 256 × 256 · final upsampling

Variant	Uses	Map for 256 × 256	Final upsampling	Result
FCN-32s	only the 1/32 map	8 × 8	×32	strong meaning, coarse boundaries
FCN-16s	adds the 1/16 map	16 × 16	×16	better spatial detail
FCN-8s	adds the 1/8 map	32 × 32	×8	even better boundaries

Lesson: FCN: turning a classifier into a segmenter

Table: method · how · good · limit

Method	How	Good	Limit
Nearest neighbour	Copy each value into an F × F block	Simple, fast, keeps the original values	Blocky boundaries, learns nothing
Bilinear	Weighted average of the 4 nearest inputs	Smooth	Only blends, does not know where a boundary lies
Transposed convolution	Learned kernel spread over the output	Learns shapes, such as a road edge against a person edge	Can still fail if the encoder lost the information

Lesson: Three ways to make a map bigger

Table: · encoder · decoder

	Encoder	Decoder
Resolution	reduces	restores
Receptive field	grows	
Produces	high level meaning	dense class scores
Answers	What is present?	Where exactly?

Lesson: Encoder, decoder and skip connections

Table: fusion · how · cost (128 × 128 map, 64 + 64 channels, 3 × 3 conv to 64)

Fusion	How	Cost (128 × 128 map, 64 + 64 channels, 3 × 3 conv to 64)
Concatenate (U-Net)	Stack both feature sets, the next conv learns how to mix them	$3 \cdot 3 \cdot 128 \cdot 64 + 64 = 73{,}792$
Add (FCN, FPN, ResNet)	Shapes must match, both go into one shared space	$3 \cdot 3 \cdot 64 \cdot 64 + 64 = 36{,}928$

Lesson: Encoder, decoder and skip connections

Table: part · what happens

Part	What happens
Contracting path	Two 3 × 3 convs, then 2 × 2 max pool. Resolution halves, channels double, features become more semantic.
Bottleneck	The most compressed map: strongest context, coarsest detail.
Expanding path	2 × 2 up-convolution, concatenate the skip, two 3 × 3 convs. A final 1 × 1 conv gives class scores.
Skips	U-Net does not wait until the end to recover detail. It injects detail at every decoder stage.

Lesson: U-Net

Table: question · answer · reason

Question	Answer	Reason
Which feature is from an early layer?	A: sharp edges, local texture, exact position	Those live at high resolution
Which is from the bottleneck?	B: object meaning, scene context, coarse layout	Deep semantics
Which should receive a skip?	C, the upsampled semantic map, receives A	C needs its boundaries refined
Why does the decoder need A and B?	A is where, B is what	Masks need local edges plus global context

Lesson: U-Net

Table: boundary problem · context problem

Boundary problem	Context problem
Edges may be thin or irregular	A grey patch could be road, wall or roof
Nearby classes may look alike	Wide scene context resolves it
The mask can leak outside the object	Deep features and large receptive fields help
High resolution features help	But enlarging the receptive field has a cost

Lesson: Dilated convolution, ASPP and DeepLab

Table: design · key idea · remember it as

Design	Key idea	Remember it as
FCN	dense class score maps	the first classifier to segmenter conversion
Encoder-decoder	compress then upsample	the general shape for pixel prediction
U-Net	skip connections at every level	strong boundaries, the medical baseline
DeepLab	dilated conv and ASPP	multi-scale context with dense features

Lesson: Dilated convolution, ASPP and DeepLab

Table: application · sensible start · why

Application	Sensible start	Why
Medical tumour boundary	U-Net or encoder-decoder	Boundary accuracy with little data
Street scenes	DeepLab style or a transformer	Many scales and much context
Drone or phone	Lightweight encoder-decoder	Latency and memory
Satellite land cover	Multi-scale context model	Large regions and very small structures

Lesson: Choosing an architecture from its failures

Table: failure · architectural fix

Failure	Architectural fix
Blurry object boundaries	Skip connections, a high resolution decoder
Small object missed	High resolution features, multi-scale design
Road confused with sidewalk	Broader context, DeepLab style modules
Thin structure disappears	Less aggressive downsampling, more skips
Object scale varies a lot	ASPP, feature pyramids

Lesson: Choosing an architecture from its failures

TIPS AND TRAPS

REMEMBER A smaller final jump means less blur. For a 512 × 512 image FCN-8s predicts on a 64 × 64 grid, four times finer in each direction than the 16 × 16 grid of FCN-32s.

Lesson: FCN: turning a classifier into a segmenter

COMMON MISTAKE When the kernel size is not divisible by the stride, some output cells receive more overlapping contributions than others, which prints a checkerboard pattern into the mask. Fixes: kernel = 2 × stride (k = 4, s = 2), or resize with nearest or bilinear and then apply a normal convolution, or use sub-pixel convolution (PixelShuffle).

Lesson: Three ways to make a map bigger

IN THE EXAM Teaching sentence from the slide: deep features say this region is probably a car, skip features decide the car boundary is exactly here.

Lesson: U-Net

IN THE EXAM

The class answered context for road against sidewalk confusion, and both slide decks agree. The recording has the lecturer saying boundary, which is most likely a transcription slip. In exams: road and sidewalk confusion is a context problem.

Lesson: Choosing an architecture from its failures

6.3 Training and evaluating segmentation

Pixel cross-entropy, weighted CE, focal and Dice losses, Dice, IoU and mIoU, the accuracy trap, augmentation, leakage and debugging.

KEY IDEAS

- **Pixel loss** L = −(1 ÷ |Ω|) Σ ln p over valid pixels. Ignore label 255 leaves both the sum and the count. P(car) 0.9, 0.5, 0.05 gives 0.105, 0.693, 2.996.
- **Imbalance** Weights w_c = N ÷ (C n_c): pedestrian 20 in the street image. Focal $(1 - p_t)^\gamma$ scales by difficulty. Soft Dice 1 − 2Σpg ÷ (Σp + Σg), usually with CE.
- **Metrics** Accuracy hides rare classes (90% with tumour IoU 0). Dice = 2TP ÷ (2TP + FP + FN), IoU = TP ÷ (TP + FP + FN), Dice = 2IoU ÷ (1 + IoU). mIoU from a dataset confusion matrix.
- **Pipeline** Same spatial transform for image and mask, colour only for the image. Masks resize with nearest neighbour. Fit statistics on the training split. Split by patient, scene or video.
- **Debugging** Six failures: boundary, small or thin, class confusion, domain shift, occlusion, over-smoothing. Check alignment, IDs, ignore handling, imbalance and per class IoU before blaming the architecture.

FORMULAS

Pixel-wise cross-entropy Exam favourite

$$L = -\frac{1}{|\Omega|} \Sigma_{i \in \Omega} \ln p_{i, y_i}$$

Ω is the set of valid pixels and p_{i, y_i} the probability given to the true class at pixel i. The one hot target picks out that one probability.

Lesson: Masks, logits and the pixel loss

Weighted cross-entropy Exam favourite

$$L = \frac{\Sigma_i w_{y_i} (-\ln p_{i, y_i})}{\Sigma_i w_{y_i}}$$

Rare or important classes get larger weights.

Lesson: Class imbalance: weighted CE, focal and Dice

Inverse frequency weight

$$w_c = \frac{N}{C \cdot n_c}$$

N pixels, C classes, n_c pixels of class c. Every class then contributes N ÷ C weighted pixels.

Lesson: Class imbalance: weighted CE, focal and Dice

Focal loss

$$\mathrm{FL} = -(1 - p_t)^\gamma \ln p_t$$

The same function as in detection (chapter 5.2). It scales by difficulty rather than by class.

Lesson: Class imbalance: weighted CE, focal and Dice

Soft Dice loss Exam favourite

$$L_{\mathrm{Dice}} = 1 - \frac{2 \Sigma p_i g_i + \varepsilon}{\Sigma p_i + \Sigma g_i + \varepsilon}$$

Uses probabilities, not 0 or 1, so it is differentiable. ε avoids 0 ÷ 0 when the class is absent.

Lesson: Class imbalance: weighted CE, focal and Dice

Dice Exam favourite

$$\mathrm{Dice} = \frac{2|P \cap G|}{|P| + |G|} = \frac{2\mathrm{TP}}{2\mathrm{TP} + \mathrm{FP} + \mathrm{FN}}$$

P prediction, G ground truth. The F1 score of the pixels.

Lesson: Metrics: accuracy, Dice, IoU and mIoU

IoU (Jaccard) Exam favourite

$$\text{IoU} = \frac{|P \cap G|}{|P \cup G|} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}$$

The same IoU as for boxes, applied to pixel sets.

Lesson: Metrics: accuracy, Dice, IoU and mIoU

Relation

$$\text{Dice} = \frac{2 \cdot \text{IoU}}{1 + \text{IoU}}, \quad \text{IoU} = \frac{\text{Dice}}{2 - \text{Dice}}$$

Dice ≥ IoU always, equal only at 0 and 1.

Lesson: Metrics: accuracy, Dice, IoU and mIoU

TABLES

Table: prediction for a car pixel · p(car) · loss –ln p

Prediction for a car pixel	P(car)	Loss –ln P
Good	0.90	0.105
Uncertain	0.50	0.693
Bad	0.05	2.996

Lesson: Masks, logits and the pixel loss

Table: class · pixels · share

Class	Pixels	Share
road	50,000	50%
sky	25,000	25%
building	20,000	20%
car	4,000	4%
pedestrian	1,000	1%

Lesson: Class imbalance: weighted CE, focal and Dice

Table: situation · loss

Situation	Loss
Standard many class problem	Cross-entropy
Rare but important classes	Weighted CE
Many easy pixels, few hard ones	Focal loss
Tiny foreground, overlap is the goal	Dice, usually with CE
Edges matter	Add a boundary loss

Lesson: Class imbalance: weighted CE, focal and Dice

Table: application · prefer · why

Application	Prefer	Why
Small brain tumour	Dice or IoU, plus recall	Tiny foreground, a miss is costly
Road and sidewalk edges	Boundary metric plus IoU	Edges matter
Balanced many class benchmark	mIoU	Every class counts
Surgical planning, lane markings	Boundary F-score, HD95	A few pixels of edge error matter

Lesson: Metrics: accuracy, Dice, IoU and mIoU

Table: split · role

Split	Role
Train	Updates the weights
Validation	Tunes settings and selects models
Test	Used once for the final report

Lesson: The training pipeline and data leakage

Table: failure · symptom · likely cause · possible fix

Failure	Symptom	Likely cause	Possible fix
Boundary errors	Right class, wrong shape, mask leaks out	Downsampling, weak decoder, rough labels	Skips, high resolution features, boundary loss
Small or thin objects	Lane marking, vessel or tiny pedestrian vanish	Aggressive downsampling	High resolution branches, multi-scale features, class weights, crops around rare objects
Class confusion	Road and sidewalk, tumour and oedema, crop and weed, river and shadow	Similar texture, weak context	Context modules, multi-scale features, better labels
Domain shift	Works on sunny days, fails on rainy nights, new scanner, new satellite sensor	Test data differs from training data	Augmentation, diverse data, domain adaptation, self supervised pre-training
Occlusion	Partly hidden objects incomplete, two objects merge	No evidence for the hidden part	Instance or panoptic methods, more context
Over-smoothing	Fine detail disappears	Too much pooling or blur, big upsampling jumps	Skips, sharper decoder, boundary loss

Lesson: Failure modes and debugging

Table: check before blaming the architecture · bug it catches

Check before blaming the architecture	Bug it catches
Are image and mask aligned after augmentation?	Flip or crop applied to only one
Are class IDs intact after resizing?	Bilinear on masks creates phantom classes
Are ignored pixels handled in the loss?	Ignore counted as a class, wrong denominator
Is the imbalance severe?	A rare class learns nothing
Is validation representative?	Leakage or an easy validation set
Which classes have low IoU?	A good mIoU hiding one dead class
Are errors mostly boundary, small object or domain?	Points to the right fix
Did you look at the masks?	Checkerboards, shifts, systematic artefacts

Lesson: Failure modes and debugging

Table: scenario · what matters · direction

Scenario	What matters	Direction
Medical	Boundaries, Dice, expert review	U-Net or nnU-Net, CE + Dice, patient level validation
Driving	Real time, small objects, weather	Efficient multi-scale model, weather augmentation, pedestrian IoU
Satellite	Huge images, scale, sensor shift	Multi-scale context, tiling, cross sensor augmentation
Phone or drone	Latency and memory	Small encoder, light decoder, lower resolution, quantisation

Lesson: Failure modes and debugging

TIPS AND TRAPS

COMMON MISTAKE Very large weights can make training unstable or raise false positives. Weights can follow importance, not only frequency.

Lesson: Class imbalance: weighted CE, focal and Dice

COMMON MISTAKE Benchmarks add one confusion matrix over the whole dataset, then compute IoU per class. Averaging per image IoUs is different and noisier, since an image without a class gives 0 ÷ 0. Say which one you report.

Lesson: Metrics: accuracy, Dice, IoU and mIoU

IN THE EXAM Leakage in segmentation also comes from slices of one patient or frames of one video landing in both train and test. Split by patient, scene or video, never by random image. A model with high training IoU and low validation IoU is probably overfitting.

Lesson: The training pipeline and data leakage

STEP 7 Vision meets language and 3D

7.1 Connecting vision and language

CLIP, contrastive learning and zero-shot prompts, transformers for segmentation, SAM, DALL-E, restoration and stereo triangulation.

KEY IDEAS

- **CLIP** Image and text encoders into one space. Cosine similarity. Symmetric contrastive loss over an N × N matrix with temperature τ. 400 million pairs, batch 32,768.
- **Zero-shot** One prompt per class, pick the highest similarity, SoftMax of 100 × cosine for probabilities. Prompt wording matters and ensembling helps. CLIP gives scores, SAM gives masks.
- **Transformers** Patch = token, $(n \div p)^2$ tokens, N^2 attention pairs: 196 → 38,416, 1,024 → 1,048,576. Windows cut the cost (64× in the Swin example). SegFormer, Mask2Former queries with Hungarian matching.
- **SAM and language** SAM: ViT encoder, prompt encoder, light decoder, SA-1B (11 M images, 1.1 B masks), class agnostic. Language-guided segmentation returns a mask for a phrase. Referring segmentation answers with a mask, VQA with text.
- **Generation, restoration, 3D** DALL-E makes images from text, CLIP matches them. Restoration and super-resolution infer plausible detail, so validate clinically. Stereo $Z = fB \div d$, error $\approx Z^2 \Delta d \div (fB)$.

FORMULAS

The processing ladder Exam favourite

$$\text{pixels} \rightarrow \text{edges and textures} \rightarrow \text{shapes} \rightarrow \text{objects} \rightarrow \text{scenes and activities}$$

Pixel changes reveal edges, edges form shapes, shapes reveal objects, and objects with their relations describe the scene.

Lesson: From pixels to meaning to words

Cosine similarity Exam favourite

$$\cos(x, y) = \frac{x \cdot y}{\|x\| \, \|y\|}$$

Between −1 and 1. It ignores length and compares direction only.

Lesson: CLIP: one space for images and text

Symmetric contrastive loss

Exam favourite

$$L = \tfrac{1}{2}\,[\,\mathrm{CE}_{\mathrm{rows}}(S \div \tau) + \mathrm{CE}_{\mathrm{columns}}(S \div \tau)\,]$$

Each row is a SoftMax over captions whose correct answer is the diagonal, and each column the same over images. τ is a learned temperature. CLIP used batches of 32,768 pairs, so each image had 32,767 negatives.

Lesson: CLIP: one space for images and text

Tokens

Exam favourite

$$N = (n \div p)^2$$

n image side, p patch side. A 224 image with 16 × 16 patches gives $14^2 = 196$ tokens of $16 \cdot 16 \cdot 3 = 768$ numbers each.

Lesson: Transformers for dense prediction

Attention pairs

$$\text{global: } N^2, \quad \text{windowed: } (N \div w^2) \cdot (w^2)^2$$

w tokens per window side.

Lesson: Transformers for dense prediction

Stereo depth

Exam favourite

$$Z = \frac{f\,B}{d}$$

f focal length in pixels, B baseline between the cameras, d disparity: how far the point shifts between the two images.

Lesson: 3D from many photographs

Depth error

$$\Delta Z \approx \frac{Z^2}{f\,B}\,\Delta d$$

Error grows with the square of distance and falls with a wider baseline.

Lesson: 3D from many photographs

TABLES

Table: model · question it answers · output

Model	Question it answers	Output
CLIP	Which text best matches this image?	Similarity scores, zero-shot labels
DALL-E	What image should be created from this text?	A new image
SAM	Which pixels belong to the object I point at?	Masks, without class names
Language-guided segmentation	Which pixels match this phrase?	A mask for the phrase

Lesson: From pixels to meaning to words

Table: training · inference

Training	Inference
Image and caption pairs go in	One image and a list of class prompts go in
Both encoders produce embeddings	Prompts are embedded once and reused
All pairwise similarities in the batch	Similarity of the image with each prompt
The contrastive loss updates both encoders	The highest similarity gives the class

Lesson: CLIP: one space for images and text

Table: building block · what it does

Building block	What it does
CLIP	Aligns whole images with text
LSeg, OpenSeg	Align per pixel embeddings with the text embedding of a class name
CLIPSeg	Segments from a text or image prompt
Grounded-SAM	A text driven detector (Grounding DINO) finds boxes, SAM turns them into masks

Lesson: Promptable and language-guided segmentation

Table: · clip · dall-e

	CLIP	DALL-E
Direction	Aligns images and text	Text to image
Main task	Matching or classification	Creating a new image
Output	Similarity or a class	A generated image
Objective	Align embeddings	Generate images conditioned on text

Lesson: Generating and restoring images

Table: generative family · idea · where in this course

Generative family	Idea	Where in this course
GAN	A generator and a discriminator compete	Chapter 5.2, the GAN story
VAE	Encode to a distribution, sample, decode	Chapter 6.2, side topic
Diffusion	Learn to remove noise step by step, usually with a U-Net denoiser	Chapter 6.2 mentions the U-Net inside
Autoregressive	Predict image tokens one after another, like text	The first DALL-E

Lesson: Generating and restoring images

TIPS AND TRAPS

COMMON MISTAKE Some models now beat the estimated human error on the ImageNet benchmark. The lecturer's caution: that is task specific. The same models can still fail under unusual conditions, distribution shift, adversarial changes, occlusion and new categories. Benchmark superiority is not general intelligence.

Lesson: From pixels to meaning to words

IN THE EXAM CLIP and SAM both take prompts but answer different questions. CLIP says what an image corresponds to and outputs scores. SAM says which pixels belong to the object and outputs masks.

Lesson: CLIP: one space for images and text

COMMON MISTAKE SAM is class agnostic. It tells you where an object is, not what it is. Pair it with a detector or CLIP when you need names.

Lesson: Promptable and language-guided segmentation

COMMON MISTAKE A visually attractive restored image can still contain wrong information. In medicine, legal evidence and inspection, invented detail is dangerous, so validate restoration on the task it serves, not on how it looks.

Lesson: Generating and restoring images

8.1 Project lab: blur, sharpening and detection

The project notebook step by step: Gaussian blur, the sharpening kernel, the results table and what it can and cannot show.

KEY IDEAS

- **The experiment** Frozen YOLO11n on COCO128 (128 images, 929 objects). Only the pixels change: baseline, 9 × 9 Gaussian blur, then the sharpening kernel. Labels are copied because nothing moves.
- **Blur** GaussianBlur with σ = 0 uses σ = 0.3 ((k − 1) ÷ 2 − 1) + 0.8 = 1.7 for k = 9. Separable, 18 multiplications instead of 81. A low pass filter that removes edges and texture, hurting small and thin objects most.
- **Sharpen** [[0, −1, 0], [−1, 5, −1], [0, −1, 0]] = identity minus Laplacian = unsharp masking with a = 4. Weights add to 1. Gain 3 − 2 cos ω along an axis. Clipped to 0 to 255, which makes halos.
- **Results** Blur: P 0.6633 → 0.6802, R 0.5894 → 0.5358, mAP50 0.6706 → 0.6176, mAP50-95 0.5026 → 0.4434. Sharpening won back about 20 percent of the recall and mAP losses and dropped P to 0.6583.
- **Limits** COCO128 overlaps the training data, 128 images is a small sample, P and R come from the max F1 confidence, and one blur level was tested. Sharpen only images known to be blurred, and consider training with blur augmentation.

FORMULAS

Gaussian weight Exam favourite

$$G(x, y) \propto e^{-(x^2 + y^2) \div (2\sigma^2)}$$

Normalised so the 81 weights add to 1. A flat area keeps its brightness.

Lesson: Stage 2: the Gaussian blur

σ chosen by OpenCV Exam favourite

$$\sigma = 0.3\,((k - 1) \div 2 - 1) + 0.8$$

k = 9 gives σ = 0.3 × 3 + 0.8 = 1.7. For odd sizes up to 7 OpenCV uses fixed binomial weights instead.

Lesson: Stage 2: the Gaussian blur

The project kernel Exam favourite

$$K = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

The identity minus the 4 neighbour Laplacian. The weights add to 5 − 4 = 1, so flat areas are unchanged.

Lesson: Stage 3: the sharpening kernel

Sharpened pixel

$$S(x, y) = 5B(x, y) - B(x + 1, y) - B(x - 1, y) - B(x, y + 1) - B(x, y - 1)$$

Then rounded and clipped to 0 to 255, because ddepth = −1 keeps the 8 bit image type.

Lesson: Stage 3: the sharpening kernel

Share of the loss won back Exam favourite

$$\frac{\text{restored} - \text{degraded}}{\text{baseline} - \text{degraded}}$$

For mAP50-95: (0.4556 − 0.4434) ÷ (0.5026 − 0.4434) = 0.0122 ÷ 0.0592 ≈ 20.6 percent.

Lesson: Reading the results

F1 at the reported point

$$\text{F1} = \frac{2\,P\,R}{P + R}$$

Original 0.6242, blur 0.5994, sharpened 0.5974. By this single number sharpening is slightly worse than blur, while by mAP it is better. Different metrics can disagree on a small effect.

Lesson: Reading the results

TABLES

Table: item · choice in the notebook

Item	Choice in the notebook
Dataset	COCO128: 128 images, 929 labelled objects, 80 classes, 2 images with no objects
Model	Pretrained YOLO11n (fused): 100 layers, 2,616,248 parameters, 6.5 GFLOPs
Training	None. The same frozen detector is evaluated three times
Evaluation	model.val at image size 640 and batch 16, reporting precision, recall, mAP50 and mAP50-95
Stage 2	cv2.GaussianBlur(image, (9, 9), 0) on every image
Stage 3	cv2.filter2D(blurred, −1, K) with K = [[0, −1, 0], [−1, 5, −1], [0, −1, 0]]

Lesson: The project brief

Table: metric · question it answers

Metric	Question it answers
Precision	Of the boxes the detector reported, what share were right?
Recall	Of the real objects, what share were found?
mAP50	Area under the precision and recall curve per class, averaged over classes, with a match needing IoU at least 0.5
mAP50-95	The same, averaged over ten IoU thresholds 0.50, 0.55 and so on up to 0.95, so loose boxes score lower

Lesson: The project brief

Table: stage · precision · recall · map50

Stage	Precision	Recall	mAP50	mAP50-95
Original	0.6633	0.5894	0.6706	0.5026
Gaussian blur	0.6802	0.5358	0.6176	0.4434
Sharpened	0.6583	0.5468	0.6277	0.4556

Lesson: Reading the results

Table: metric · change · relative change

Metric	Change	Relative change
Precision	+0.0169	+2.5 percent
Recall	−0.0536	−9.1 percent
mAP50	−0.0530	−7.9 percent
mAP50-95	−0.0592	−11.8 percent

Lesson: Reading the results

Table: experiment · what it would show

Experiment	What it would show
Blur sweep: kernel 3, 5, 9, 15	How fast accuracy falls with blur, and where sharpening stops helping
Unsharp mask with tuned amount and radius	Whether a gentler or wider boost helps more than the fixed kernel
Wiener deconvolution with the known Gaussian	The best a linear method can do
Results by object size	Whether small objects carry the loss, as the examples suggest
Fine tune with blur augmentation	Whether training is a better fix than preprocessing
Evaluate on val2017	Scores free from training overlap

Lesson: How sure can we be, and what to try next

TIPS AND TRAPS

REMEMBER The labels are copied unchanged on purpose. If you resized or cropped the images, the boxes would have to change too. Blur and sharpening keep every pixel in place.

Lesson: The project brief

IN THE EXAM Your analysis found exactly this. In 000000000531.jpg detections fell from 10 to 4 and the tennis racket and distant players disappeared. In 000000000260.jpg detections fell from 8 to 3. Blur also caused class confusion: in 000000000049.jpg a horse was labelled as a dog with score 0.30.

Lesson: Stage 2: the Gaussian blur

COMMON MISTAKE filter2D computes correlation, not convolution, so the kernel is not flipped. For symmetric kernels such as this one the two are identical. The border default reflects the image without repeating the edge pixel (BORDER_REFLECT_101).

Lesson: Stage 3: the sharpening kernel

IN THE EXAM A careful conclusion uses several metrics and says which one decides. mAP summarises every threshold, so it is the main score for detectors. P and R describe one operating point.

Lesson: Reading the results

STEP 9 Exam revision

9.1 Exam revision: from school to research universities

Exam style questions from school level to research universities, with worked answers, a formula sheet and a mock quiz.

FORMULAS

Image size in bytes — Exam favourite

$$\text{width} \times \text{height} \times \text{channels} \times \text{bytes per value}$$

An 8 bit value takes 1 byte.

Lesson: Tier 1: school and first year

Colour to grey — Exam favourite

$$Y = 0.299\,R + 0.587\,G + 0.114\,B$$

Green counts most because the eye is most sensitive to it.

Lesson: Tier 1: school and first year

Negative and brightness

$$s = 255 - r, \quad s = \text{clip}(r + b,\ 0,\ 255)$$

Results outside 0 to 255 are clipped.

Lesson: Tier 1: school and first year

TABLES

Table: tier · typical format · what earns the marks · where to practise

Tier	Typical format	What earns the marks	Where to practise
School and first year	Short answers, small calculations, labelled diagrams	Correct definitions and units, careful arithmetic	Lesson 2
University semester and IIT course exams	Numericals by hand, short notes, compare and contrast	Every step shown, formula stated first, final answer boxed	Lesson 3
NPTEL and GATE style papers	Multiple choice, multiple select and numerical answer type, no partial credit	Speed, recognising the trap option, exact rounding	Lesson 4
MIT, Stanford and Cambridge style	Derivations, proofs, design questions, critique of an experiment	Clear assumptions, a complete argument, trade offs named	Lesson 5

Lesson: How computer vision is examined

Table: shortcut · use it for

Shortcut	Use it for
Same padding: p = (k − 1) ÷ 2 at stride 1	Size unchanged after a k × k conv
Each stride 2 halves the size	224 → 112 → 56 → 28 → 14 → 7
Two 3 × 3 = reach 5, three 3 × 3 = reach 7	Receptive field without a formula
Dilated kernel reach d (k − 1) + 1	3 × 3 with d = 2 reaches 5
Transposed conv out = (n − 1) s − 2p + k	k 4, s 2, p 1 exactly doubles the size
Dice = 2 IoU ÷ (1 + IoU)	Convert between the two segmentation scores

Lesson: Tier 3: multiple choice and numerical answer papers

Table: topic · formula

Topic	Formula
Grey value	$Y = 0.299\ R + 0.587\ G + 0.114\ B$
Equalisation	$s_k = \text{round}((L - 1) \times \text{CDF}(r_k))$
Gradient	$\lvert\nabla I\rvert = \sqrt{(G_x^2 + G_y^2)},\ \theta = \text{atan2}(G_y, G_x)$
Harris	$R = \det M - k\ (\text{trace}\ M)^2$
Flow constraint	$I_x\ u + I_y\ v + I_t = 0$
Conv output	$\lfloor (n + 2p - k) \div s \rfloor + 1$
Conv weights	$k^2\ C_{in}\ C_{out} + C_{out}$
Receptive field	$r \leftarrow r + (k - 1)\ j,\ j \leftarrow j\ s$
Transposed conv	$(n - 1)\ s - 2p + k$
IoU, Dice	$A \cap B \div A \cup B,\quad 2\lvert A \cap B\rvert \div (\lvert A\rvert + \lvert B\rvert)$
Precision, recall, F1	$TP \div (TP + FP),\ TP \div (TP + FN),\ 2PR \div (P + R)$
Cross entropy, focal	$-\ln p,\quad -(1 - p)^{\gamma} \ln p$
Stereo	$Z = f\ B \div d,\ \Delta Z \approx Z^2\ \Delta d \div (f\ B)$
ViT tokens	$(n \div p)^2$, attention pairs N^2

Lesson: Formula sheet and last minute checks

TIPS AND TRAPS

IN THE EXAM In a numerical, write the formula, substitute, then compute. Most lost marks come from skipping the first line, so the examiner cannot give method marks when the arithmetic slips.

Lesson: How computer vision is examined

COMMON MISTAKE A common slip is mixing bits and bytes. 24 bits per pixel is 3 bytes per pixel.

Lesson: Tier 1: school and first year

COMMON MISTAKE Write the box format before computing IoU. (x1, y1, x2, y2) and (x, y, w, h) give very different answers from the same four numbers.

Lesson: Tier 2: university semester and IIT course exams

IN THE EXAM Trap options to expect: forgetting the bias in a parameter count, using k instead of k − 1 in the receptive field, counting the background class in mIoU when the question excludes it, and rounding in the middle of a calculation.

Lesson: Tier 3: multiple choice and numerical answer papers

IN THE EXAM In derivations, name each assumption when you use it: brightness constancy, small motion, rectified cameras, known σ. Examiners often ask what breaks when an assumption fails.

Lesson: Tier 4: derivations, design and critique